IS6400 Business Data Analytics

Dr. LIU Junming

City University of Hong Kong

Lecture 5

- Quick Review
- Introduction to Clustering
- Kmeans Clustering
- Hierarchical Clustering
- Density-based Clustering

Reading Material:

- DM Chapter 7,8

Introduction

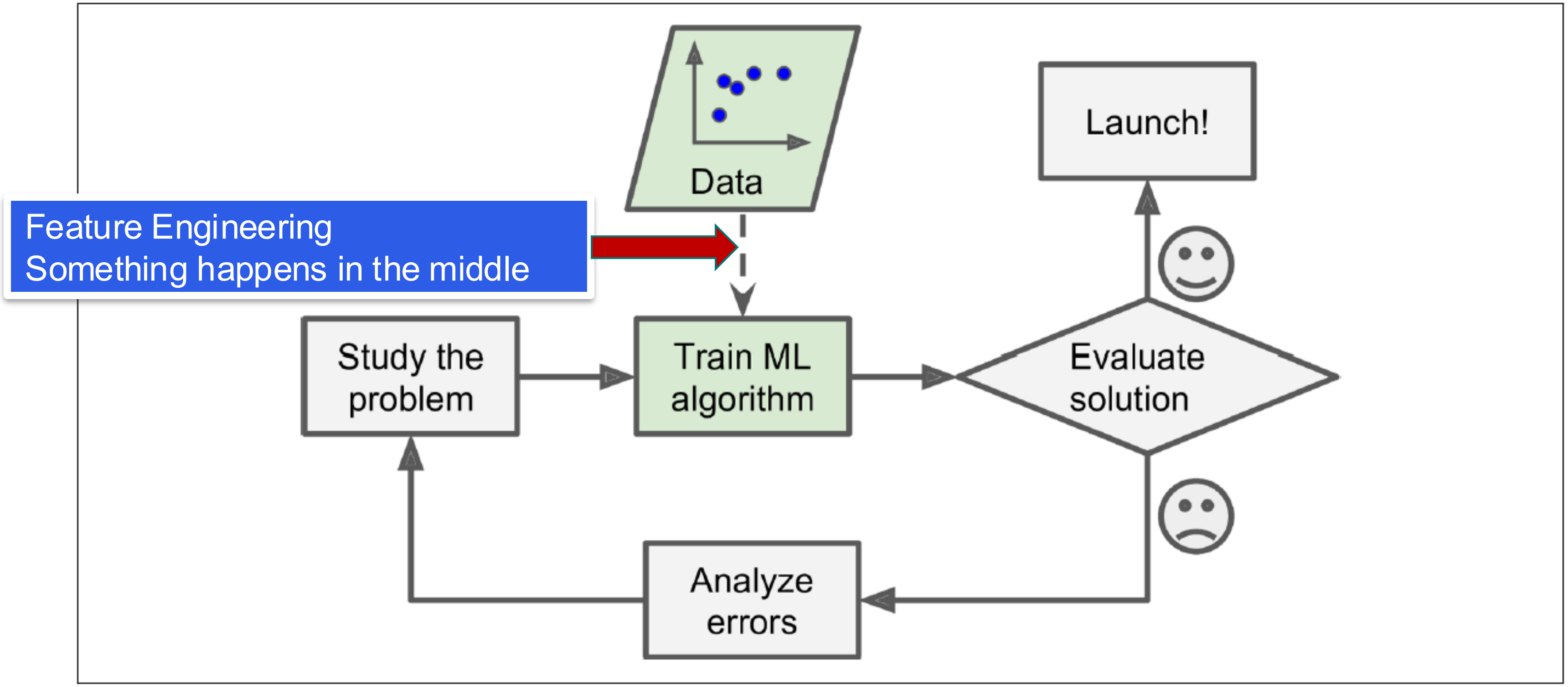
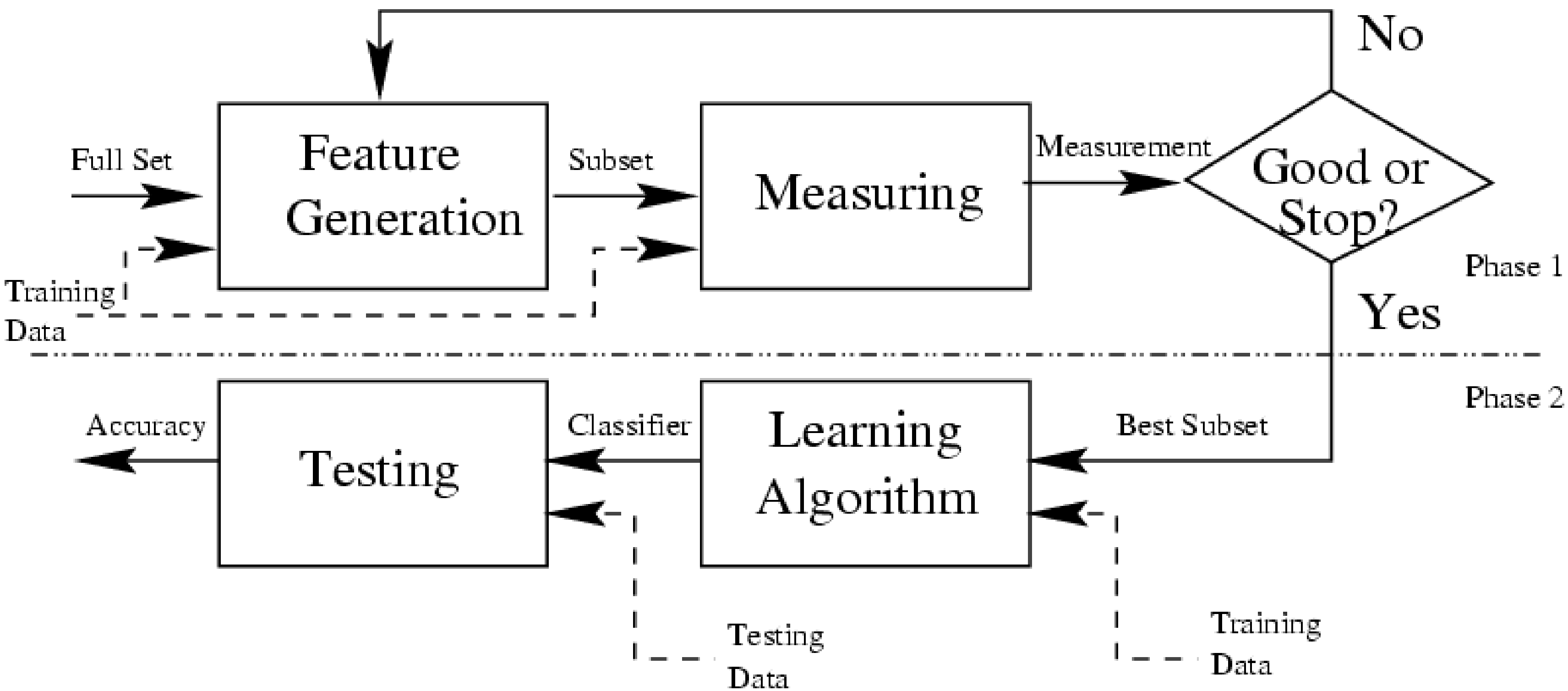


Figure 1-2. Machine Learning approach

Evaluation Measures for Ranking and Selecting Features

- The goodness of a feature/feature subset is dependent on measures
- Various measures
  - Information measures (Yu & Liu 2004, Jebara & Jaakkola 2000)
  - Distance measures (Robnik & Kononenko 03, Pudil & Novovicov 98)
  - Dependence measures (Hall 2000, Modrzejewski 1993)
  - Consistency measures (Almuallim & Dietterich 94, Dash & Liu 03)
  - Accuracy measures (Dash & Liu 2000, Kohavi&John 1997)

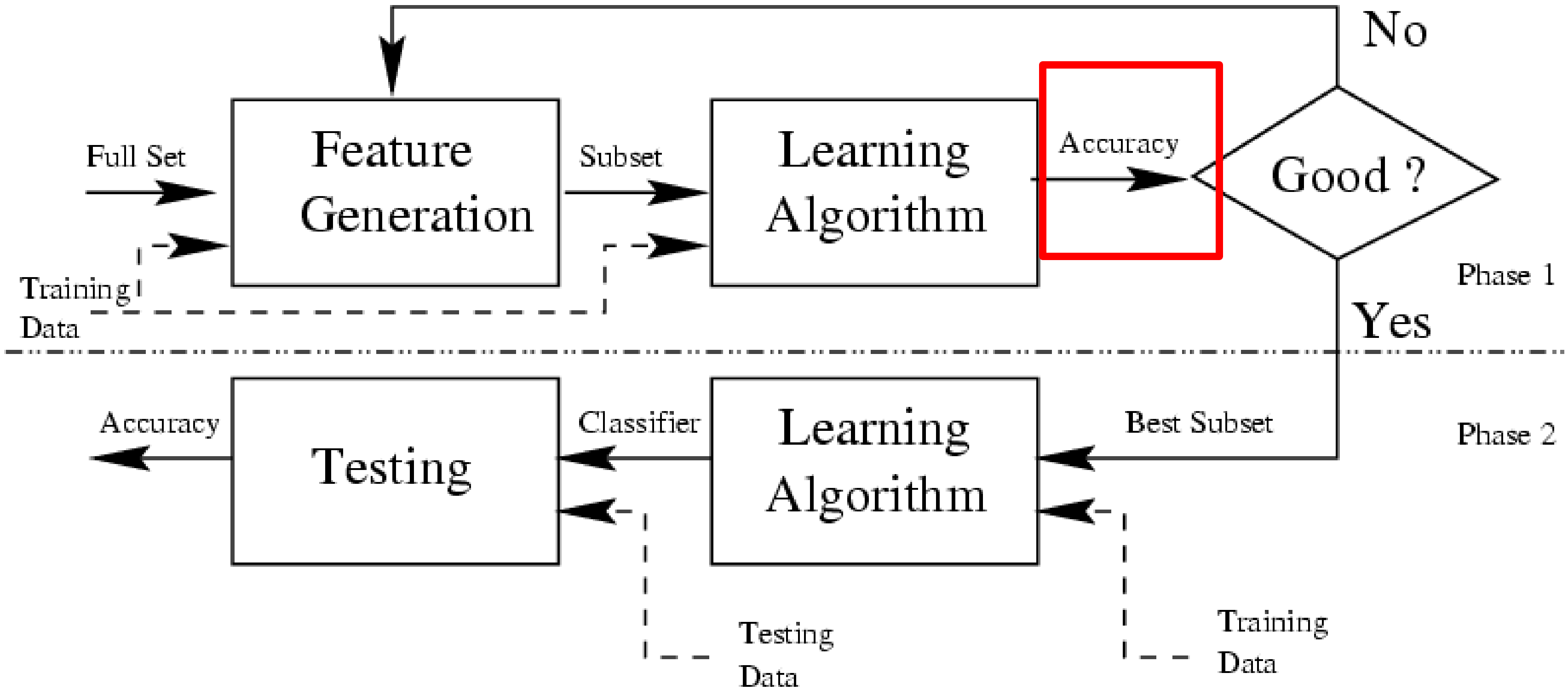
Filter Model



Feature importance measurement?

Gini index and Entropy

Wrapper Model



use the model performance as feature importance score

Feature Reduction vs. Feature Selection

- Feature reduction
  - All original features are used
  - The transformed features are linear combinations of the original features
- Feature selection
  - Only a subset of the original features are selected
- Continuous versus discrete

Recursive Feature Selection

- Can be implemented on Filter Model and Wrapper Model

Full set: {F1,F2,F3,F4}

Starting from an Empty Data Set {}

Evaluate the Feature Importance (filter model measurement or wrapper model evaluation)

Select the best feature and add it to the empty set

{} - best({F1},{F2},{F3},{F4}) - {F2}

Starting from {F2}

Evaluate the Feature Importance by Adding one feature to the set (evaluate {F2, F1}, {F2,F3},{F2,F4})

Select the best feature set and add the new feature

{F2} - best({F2, F1}, {F2,F3},{F2,F4}) - {F2,F3}

Starting from {F2, F3}

Evaluate the Feature Importance by Adding one feature to the set (evaluate {F2, F3, F1}, {F2,F3,F4})

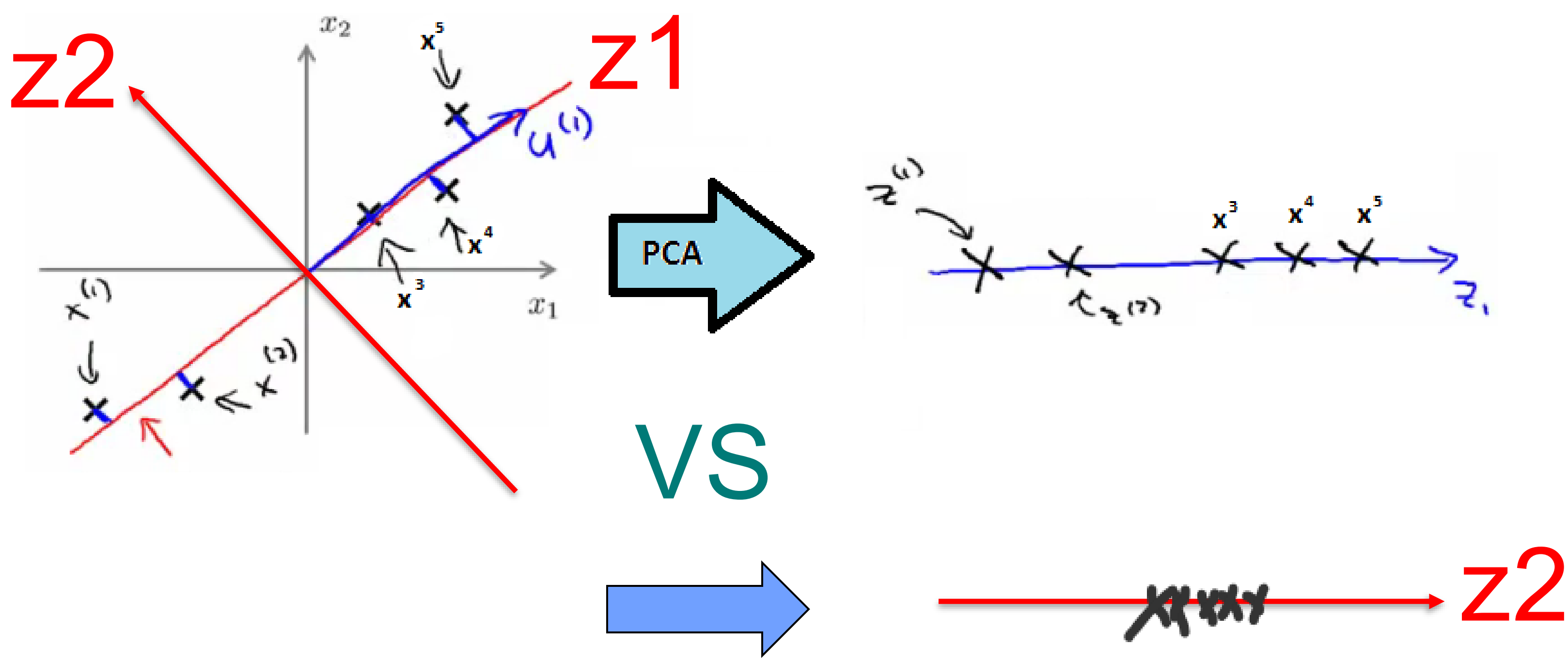
Select the best feature set and add the new feature

{F2,F3} - best({F2, F3, F1}, {F2,F3, F4}) – {F2,F3,F1}

Evaluate {F2}, {F2,F3},{F2,F3,F1},{F2,F3,F1,F4}

PCA – Importance of Principle Component

z vectors are the new, lower dimensionality feature vectors



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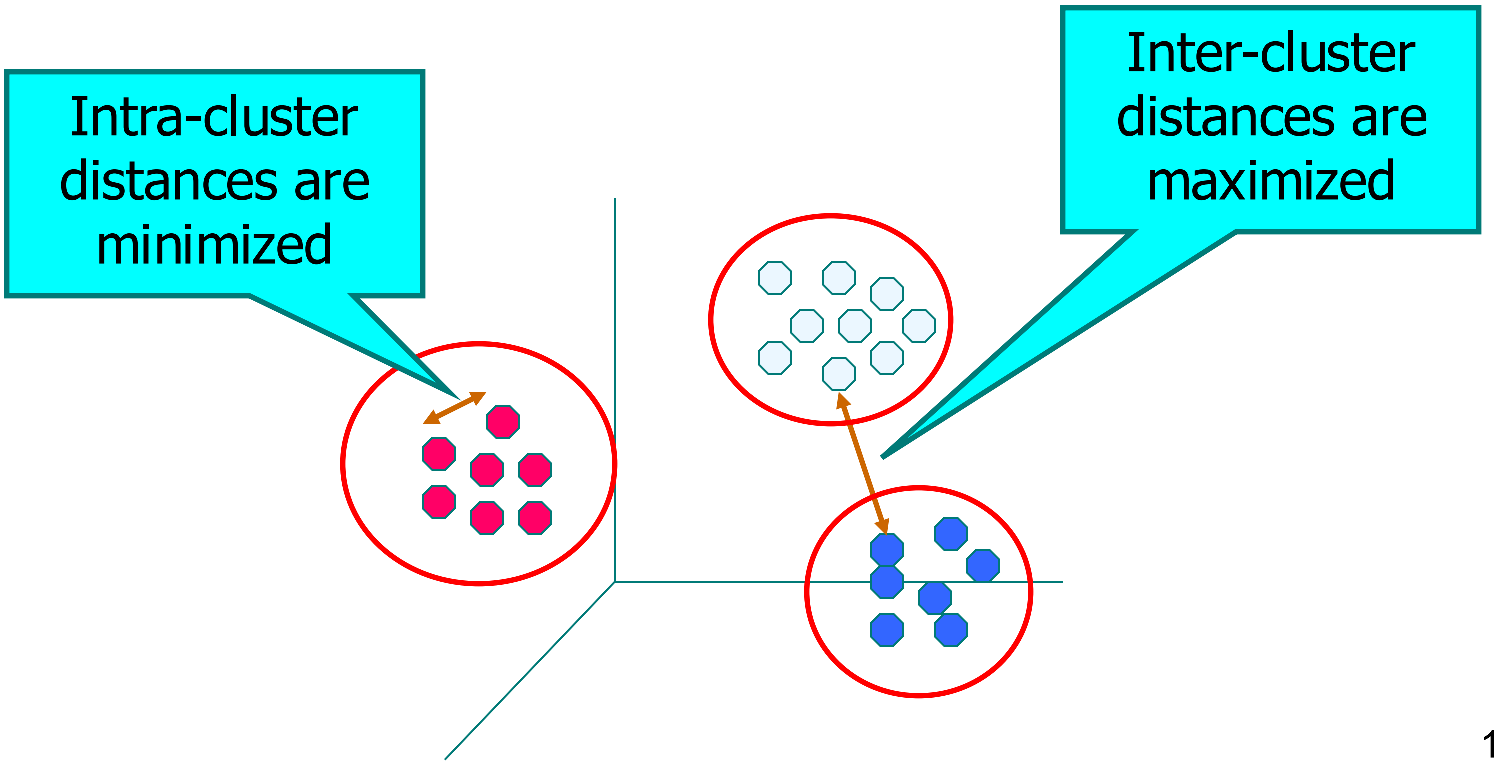
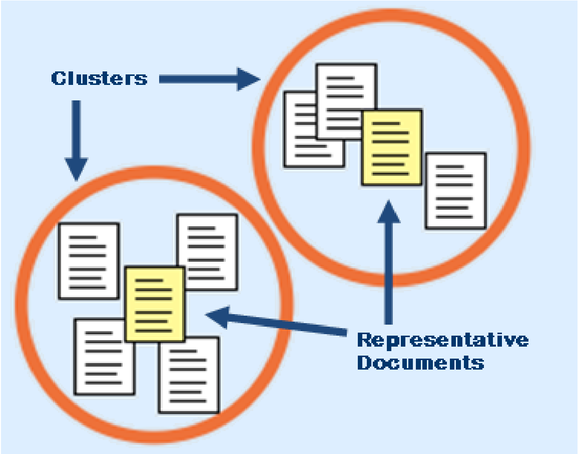
What is Cluster Analysis?

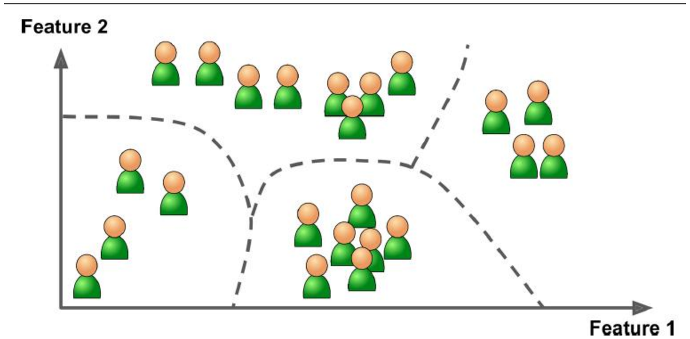
- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

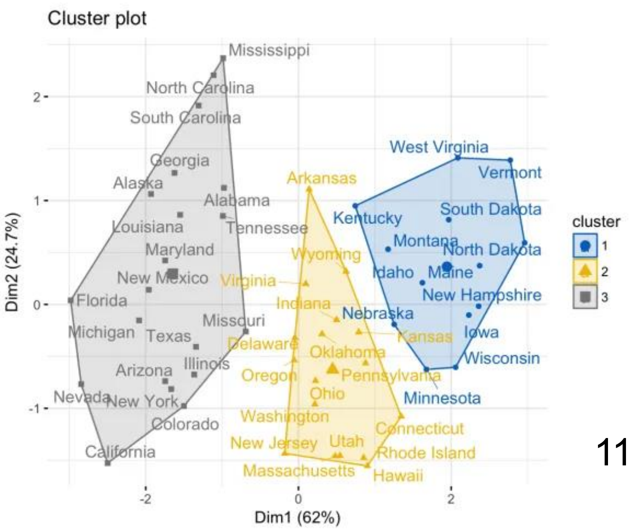


Applications of Cluster Analysis

- **Understanding**
  - Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations
- **Summarization**
  - Reduce the size of large data sets







Notion of a Cluster can be Ambiguous

How many clusters?

Six Clusters

Two Clusters

Four Clusters

Types of Clusterings

- Important distinction between partitional, hierarchical, and density-based sets of clusters
  - **Partitional Clustering**
    - A division of data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
  - **Hierarchical clustering**
    - A set of nested clusters organized as a hierarchical tree
  - **Density-based Clustering**
    - a cluster in a data space is a contiguous region separated as regions of high point density VS regions of low point density

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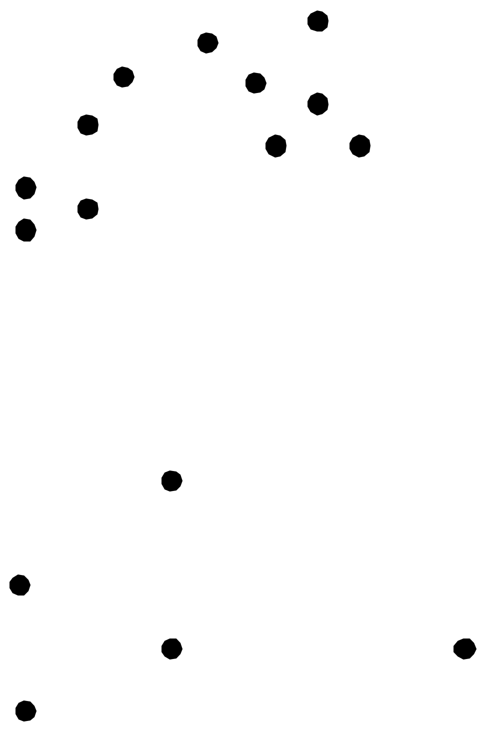
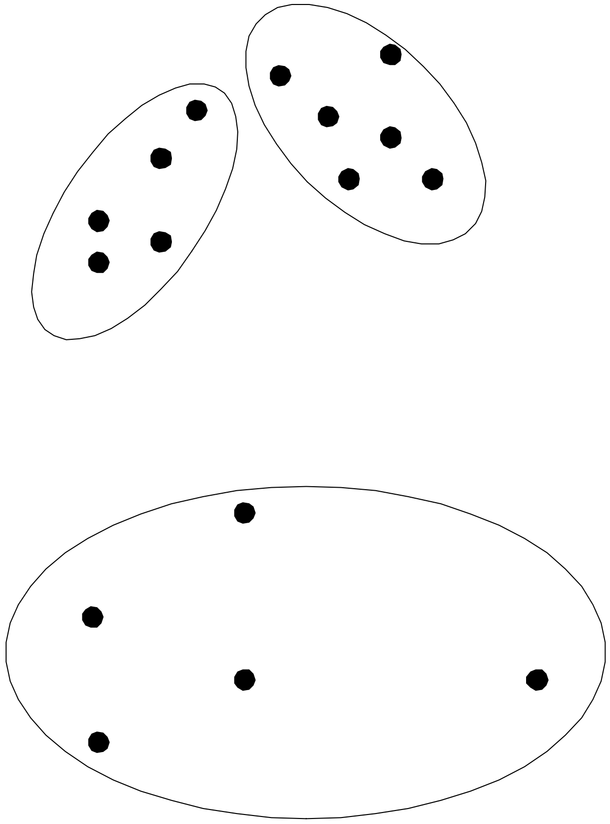
Partitional Clustering

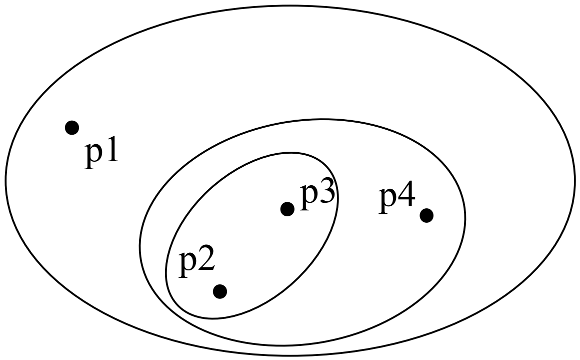


Original Points

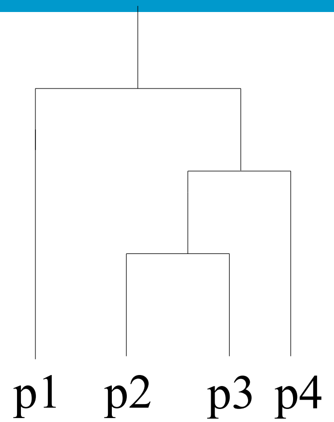


A Partitional Clustering

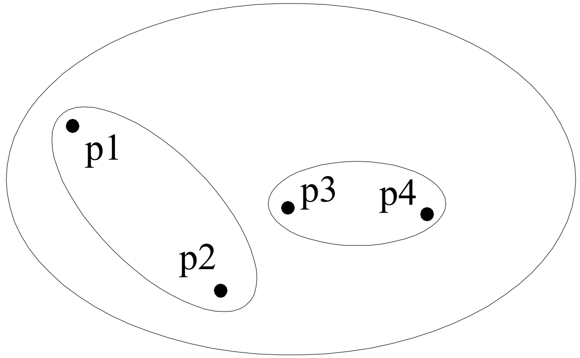
Hierarchical Clustering



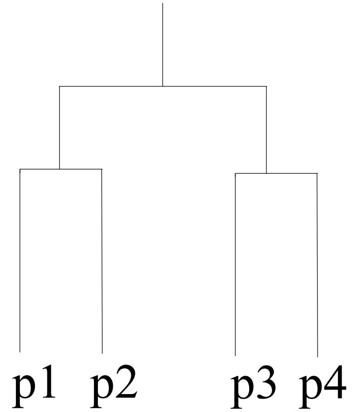
Traditional Hierarchical Clustering



Traditional Dendrogram

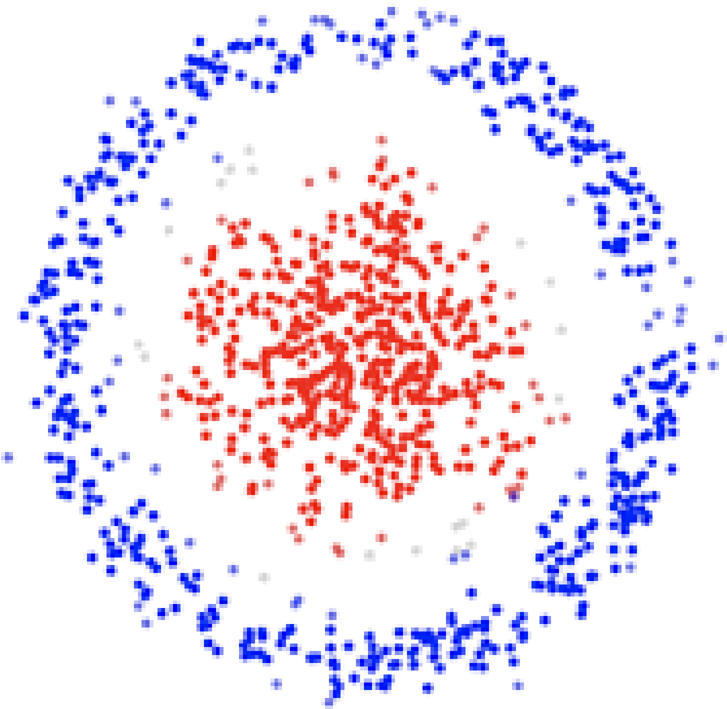


Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

Density-based Clustering



Density-based Clustering:
contiguous regions with different densities

Characteristics of the Input Data Are Important

- Type of proximity or density measure (how to measure similarities between objects)
  - Central to clustering
  - Depends on data and application
- Data characteristics that affect proximity and/or density are
  - Dimensionality
    - Sparseness
  - Attribute type
  - Special relationships in the data
    - For example, autocorrelation
  - Distribution of the data
- Noise and Outliers
  - Often interfere with the operation of the clustering algorithm

Then, Use A Similarity Metric

- Similarity expressed in terms of a distance function *d(x,y)*
- One example is Euclidian distance

$$d(x,y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Measuring Similarity Using Euclidian Distance

Object	RGB 1	RGB 2	RGB 3
Apple	220	156	140
Orange	243	194	113

$$d(obj1, obj2) = \sqrt{(220-243)^2+(156-194)^2+(140-113)^2}$$

Attribute Normalization

Homer	Mr. Burns
Age: 62	Age: 65
Income: \$990,000	Income: \$1,000,000
# of credit cards: 8	# of credit cards: 10

$$d(Homer, Burns) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

$$= \sqrt{(62-65)^2 + \underline{(990{,}000 - 1{,}000{,}000)}^2 + (8-10)^2}$$

$$\approx 10{,}000$$

Standard score	$\frac{X-\mu}{\sigma}$

Min-max feature scaling	$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$

Basic Clustering Algorithms

- K-means and its variants
- Hierarchical clustering
- Density-based clustering

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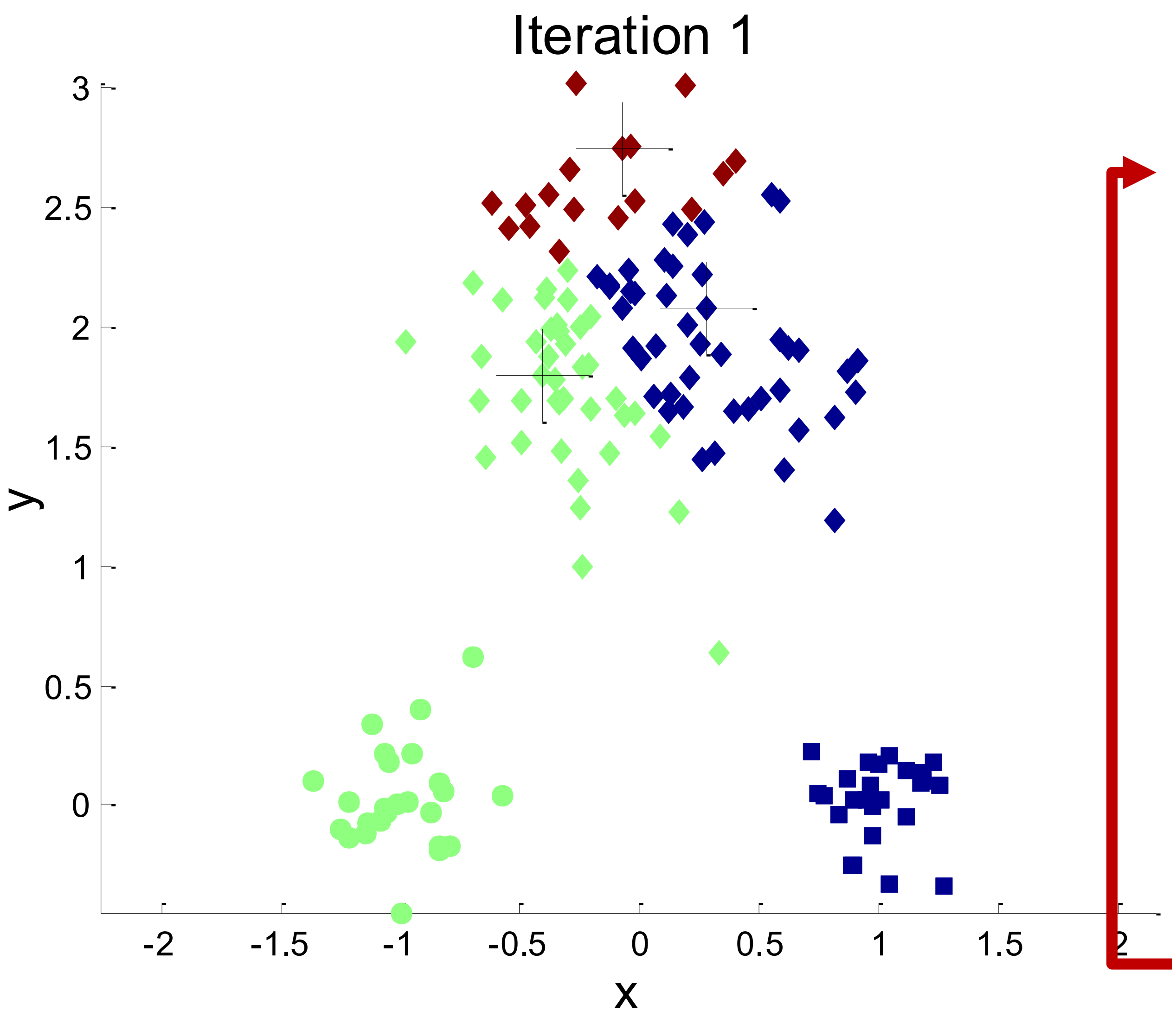
K-means Clustering

- Partitional clustering approach
- Number of clusters, K, must be specified
- Each cluster is associated with a centroid (center point)
- Each point is assigned to the cluster with the closest centroid
- The basic algorithm is very simple

```
Select K points as the initial centroids.
repeat
    Form K clusters by assigning all points to the closest centroid.
    Recompute the centroid of each cluster.
until The centroids don't change
```

Example of K-means Clustering



Initial 3 centers

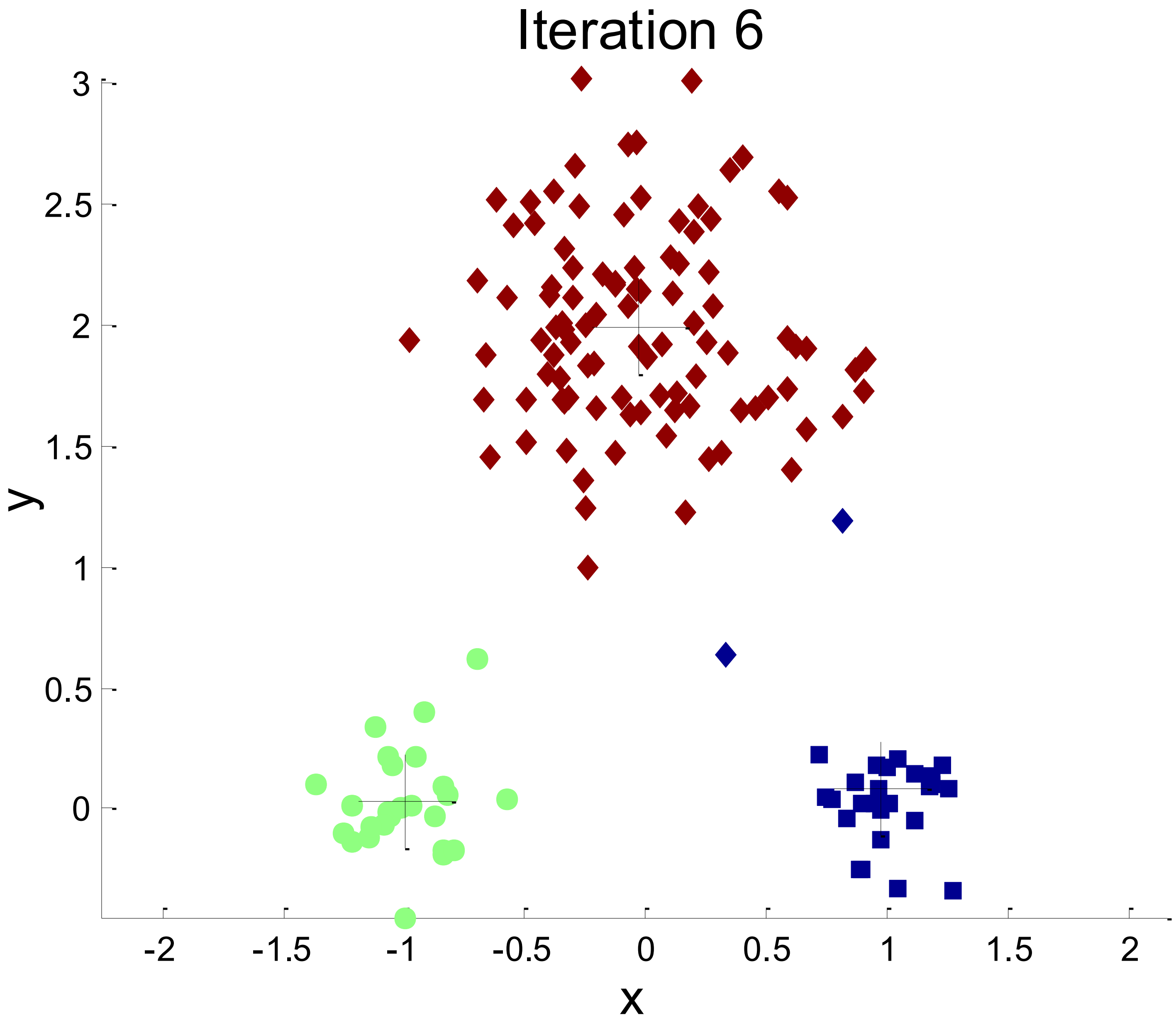
Current updated centers

Calculate distances from objects to the centers

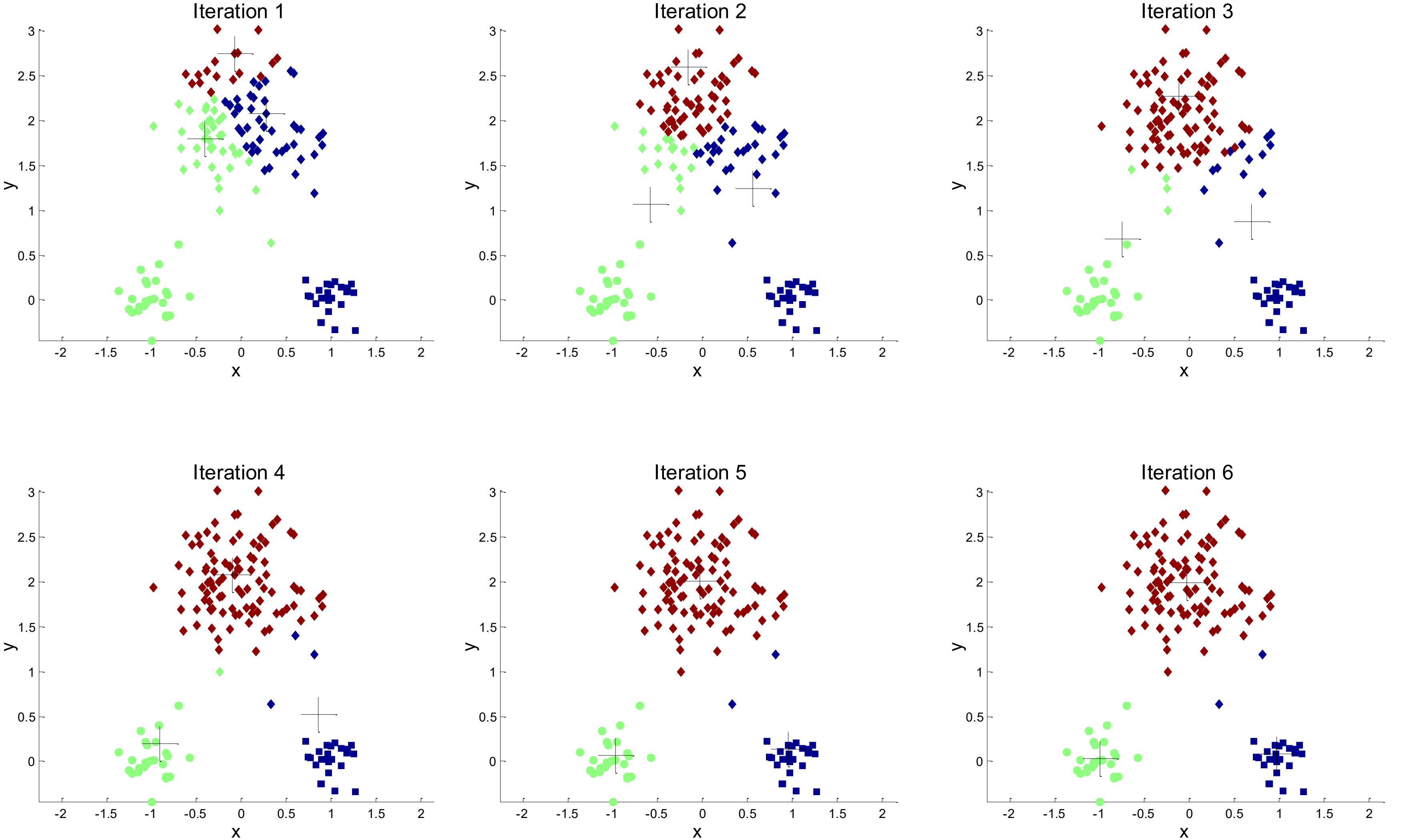
Assign objects to their closest center

According to clusters, update cluster centers

Example of K-means Clustering

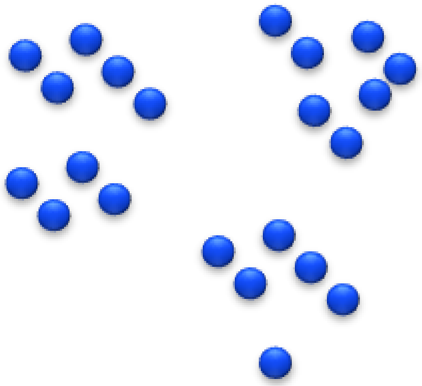


Example of K-means Clustering



Example: post office location

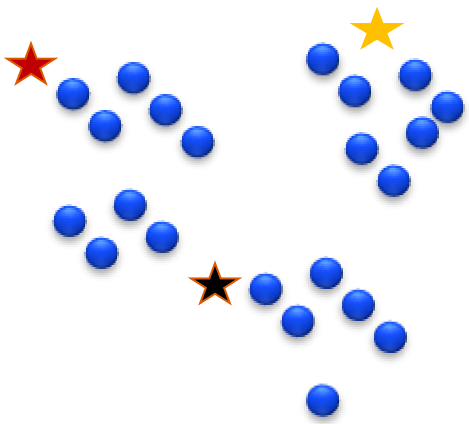
Problem: we want to decide the locations of three post offices to serve the residents.



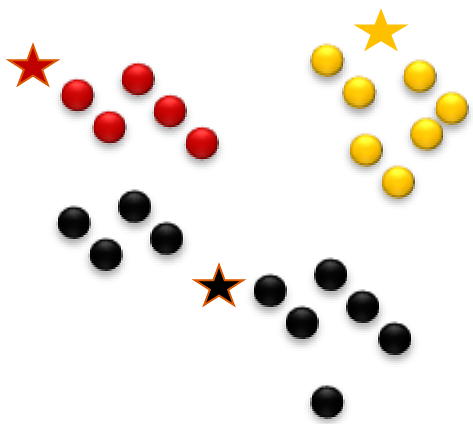
Residents locations

Example: post office location

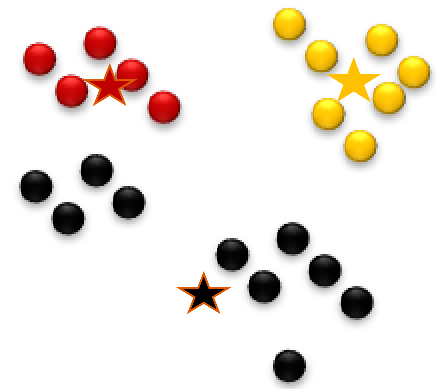
1: randomly pick three locations



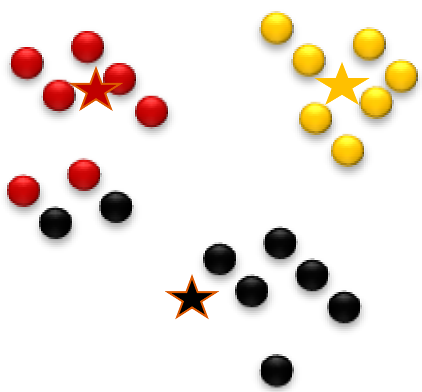
1-1: decide service region



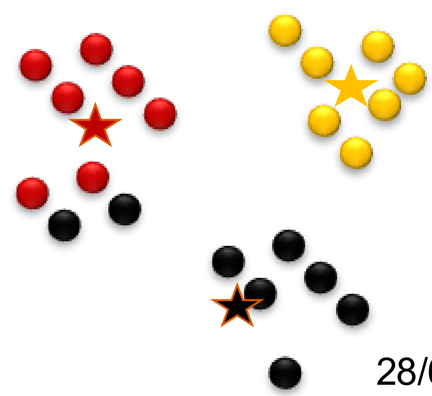
1-2: optimize locations



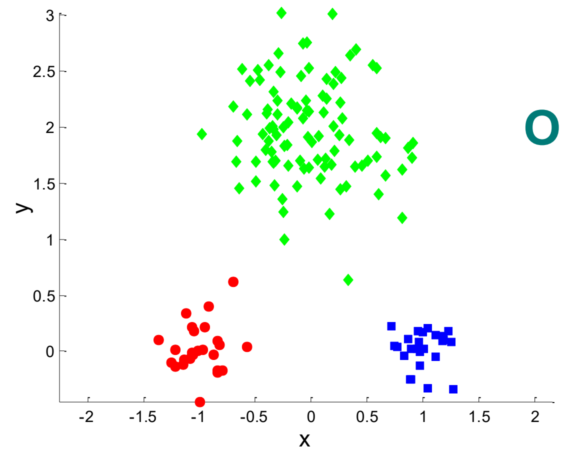
2-1: decide service region



2-2: Optimize locations



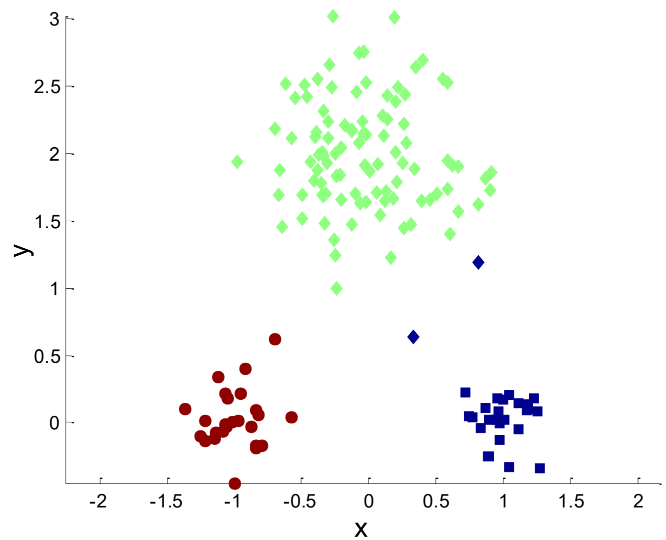
Two different K-means Clusterings



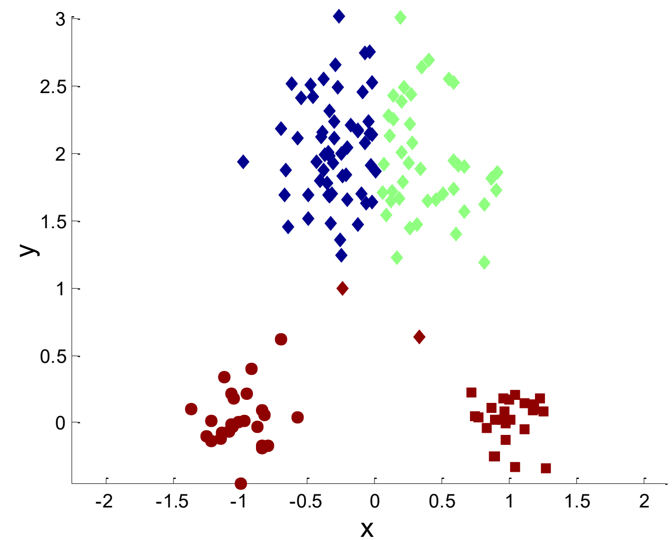
Original Points

Best initial centroids

Other initial centroid



Optimal Clustering



Sub-optimal Clustering

K-means ++

K-means++ algorithm

K-means++ enhances K-means by smartly picking initial centroids.

- Core idea: starts with one random centroid, then chooses others based on their distance from existing ones, spreading them out.
- improves both the speed and the accuracy of k-means.

- Step 1. choose first center c1
- Step 2. Choose the next center c2 from all possible points with probability

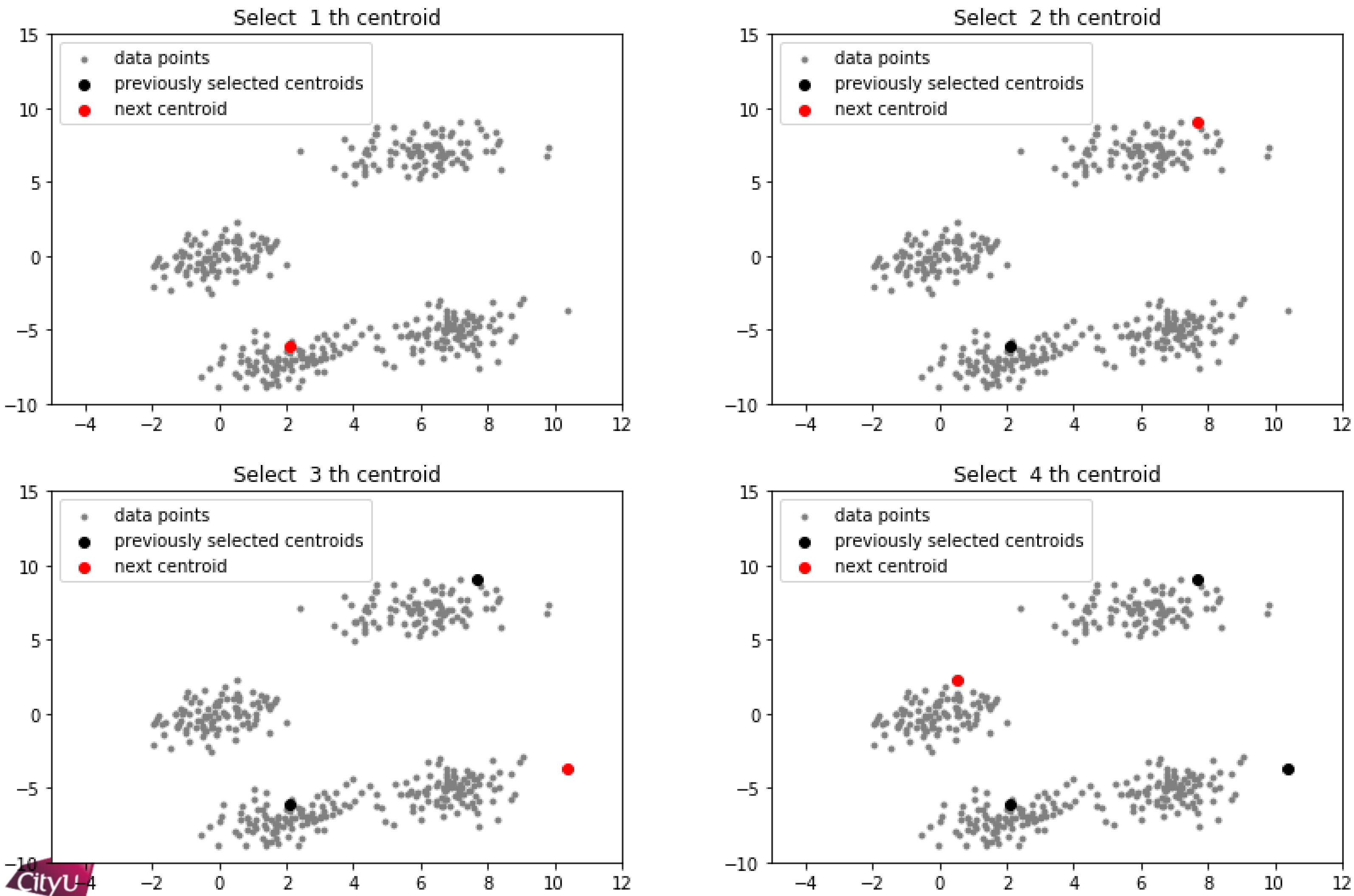
$$\frac{D(x')^2}{\sum_{x \in \mathcal{X}} D(x)^2}$$

- D(x) denote the shortest distance from a data point x to the closest center we have

reference:

Selects initial cluster centers for k-mean clustering in a smart way to speed up convergence. 2007

K-means ++



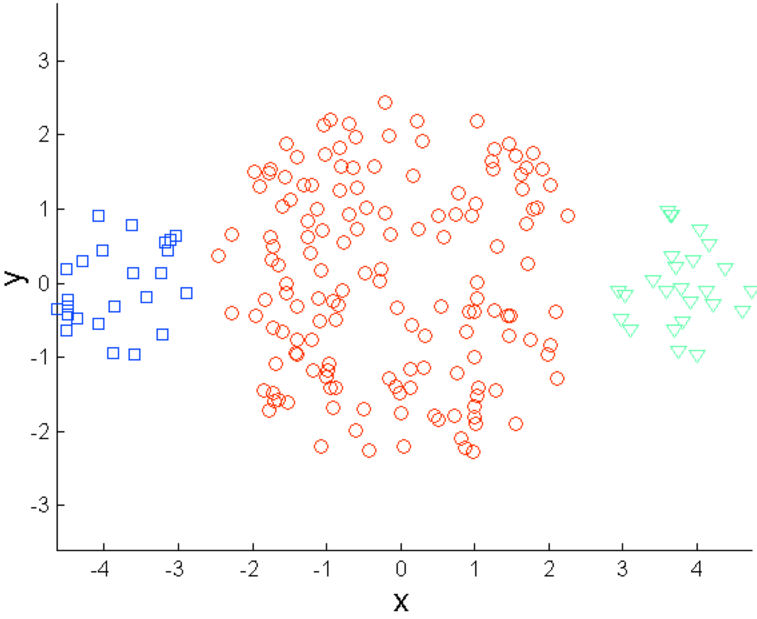
Select 1 th centroid
data points
previously selected centroids
next centroid
Select 2 th centroid
data points
previously selected centroids
next centroid
Select 3 th centroid
data points
previously selected centroids
next centroid
Select 4 th centroid
data points
previously selected centroids
next centroid

Limitations of K-means

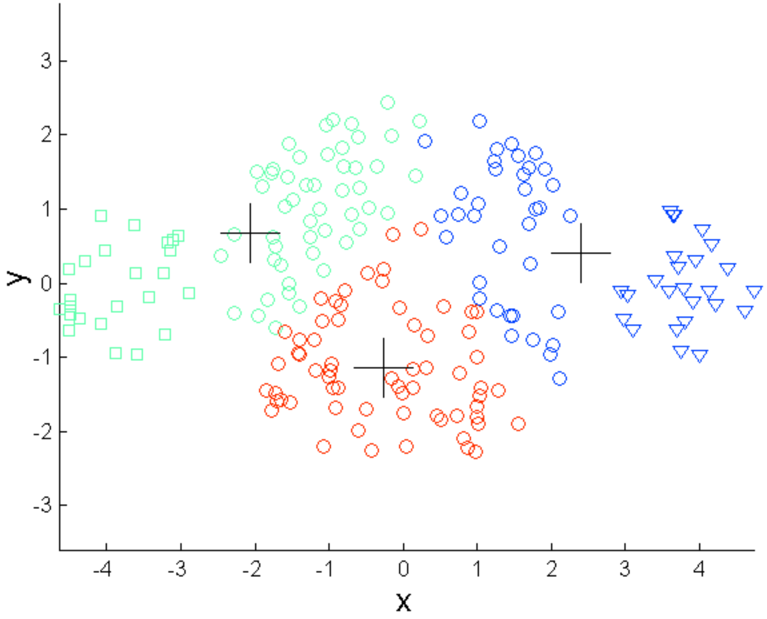
- Sub-optimal results
- K-means has problems when clusters are of differing
  - Sizes
  - Densities
  - Non-globular shapes

- K-means has problems when the data contains outliers.

Limitations of K-means: Differing Sizes

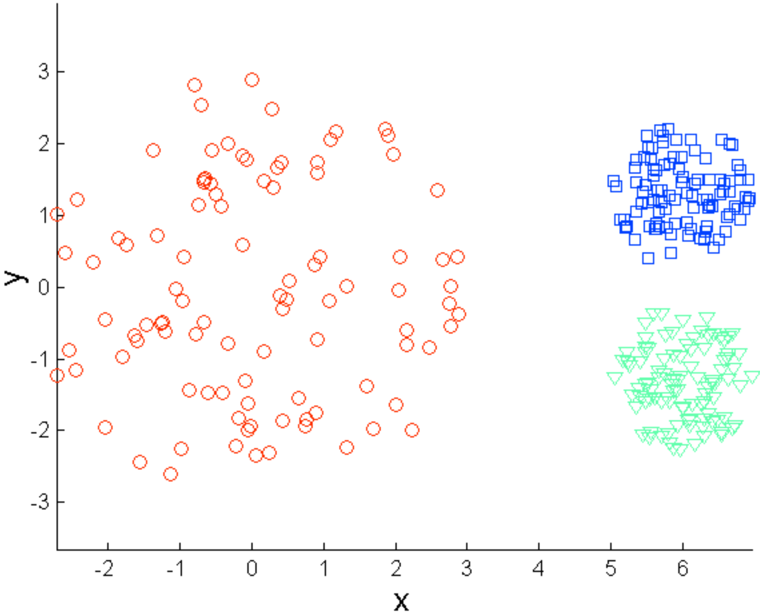


Original Points

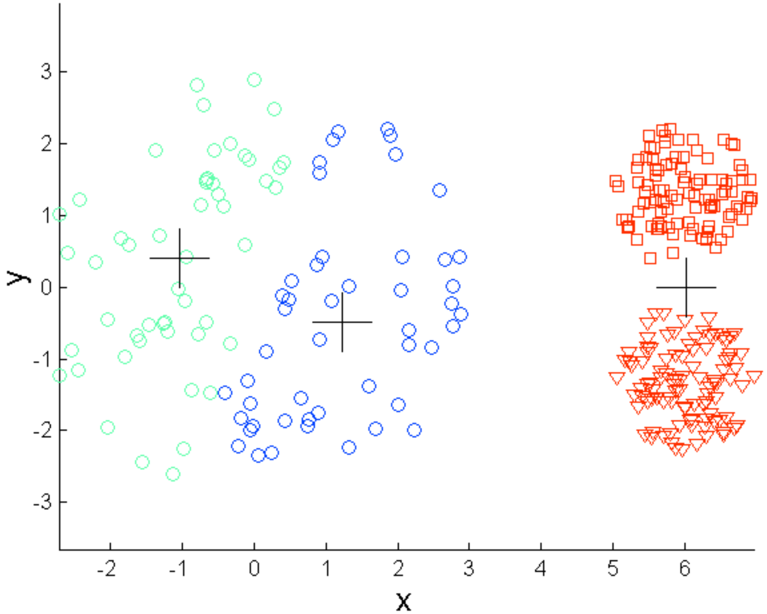


K-means (3 Clusters)

Limitations of K-means: Differing Density

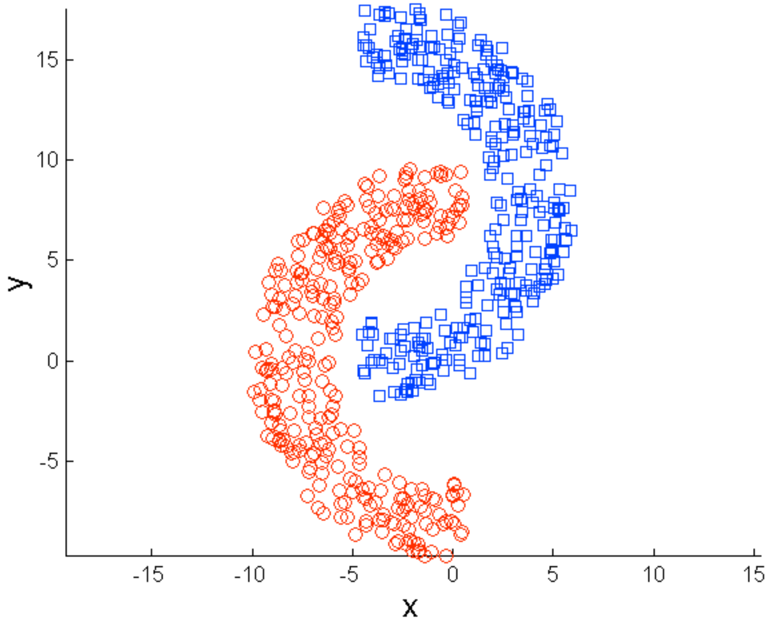


Original Points

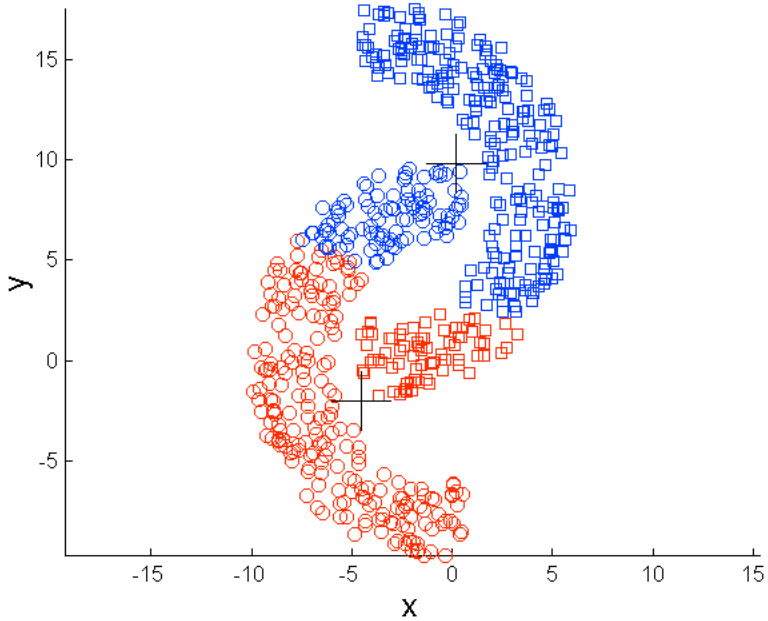


K-means (3 Clusters)

Limitations of K-means: Non-globular Shapes

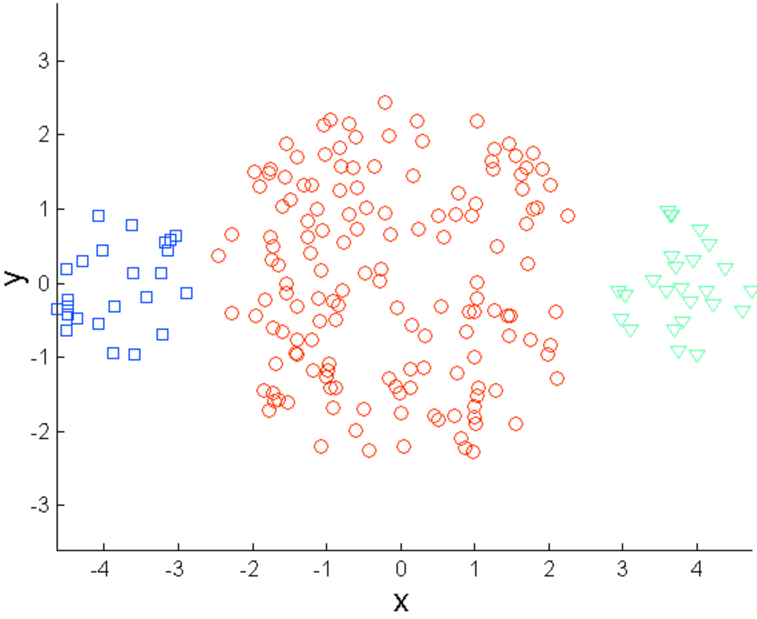


Original Points

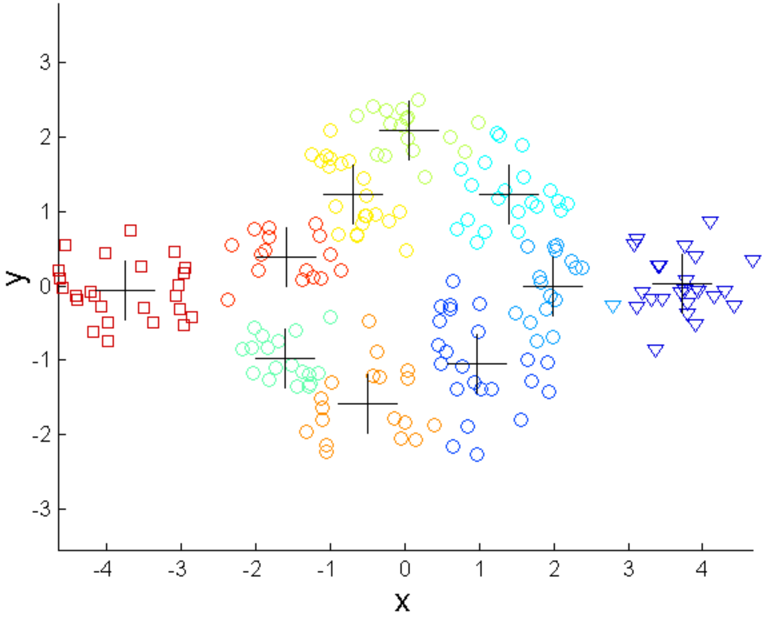


K-means (2 Clusters)

Overcoming K-means Limitations



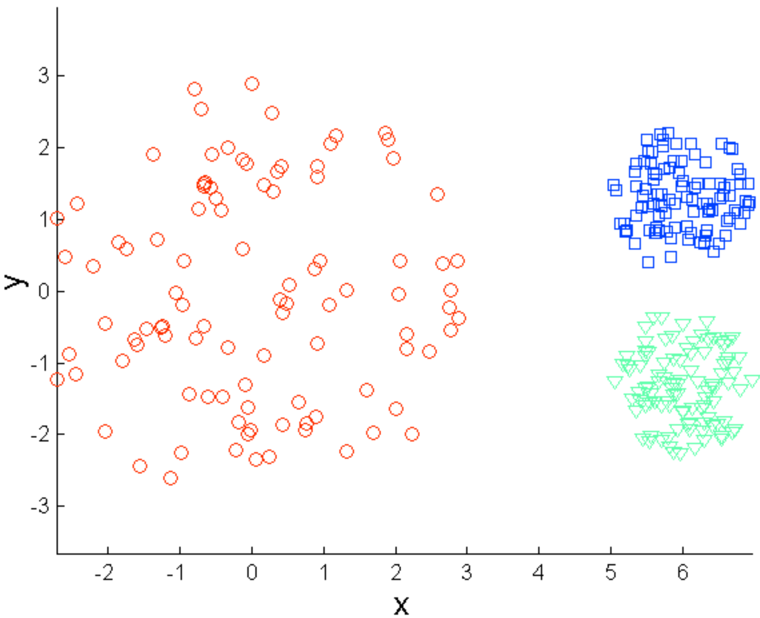
Original Points



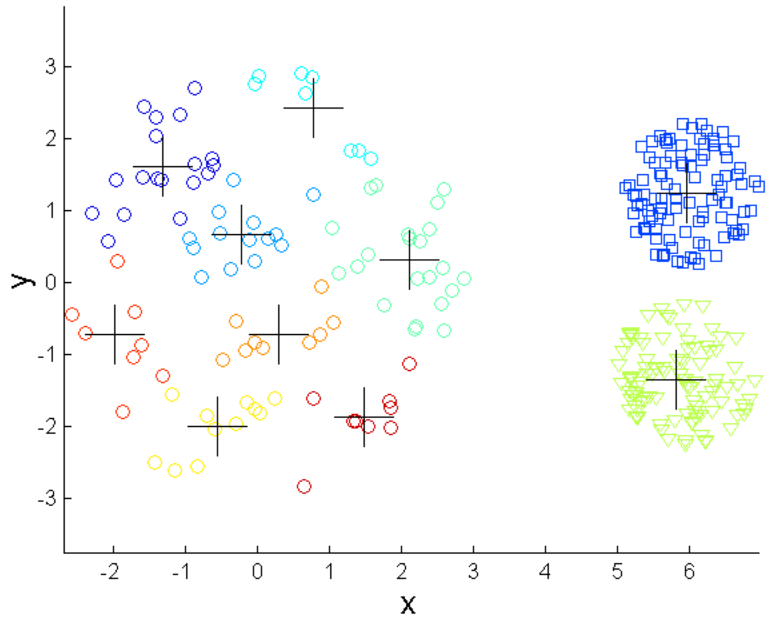
K-means Clusters

One solution is to use many clusters.
Find parts of clusters, but need to put together.

Overcoming K-means Limitations

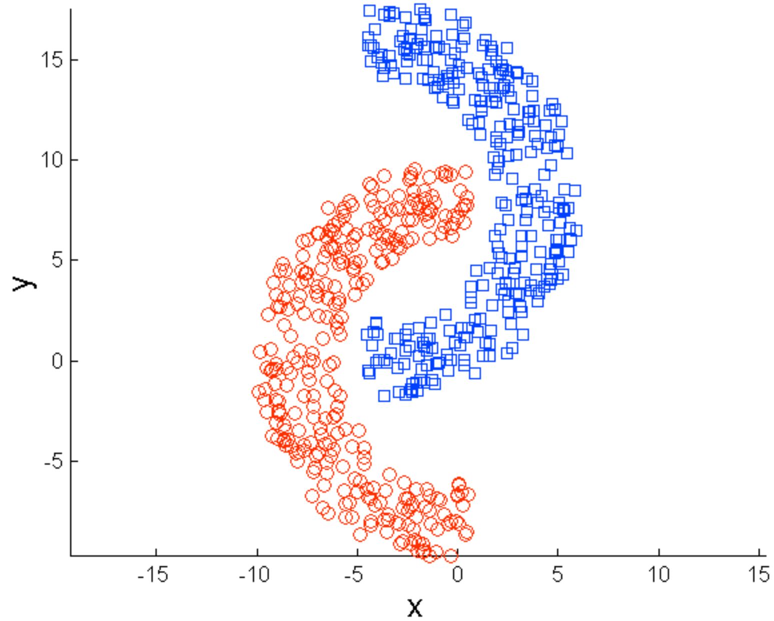


Original Points

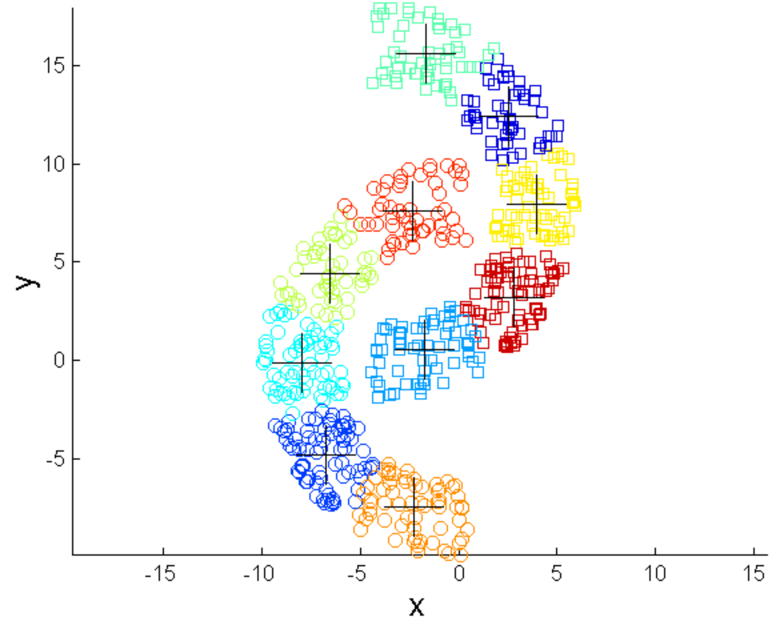


K-means Clusters

Overcoming K-means Limitations



Original Points



K-means Clusters

Bisecting K-means

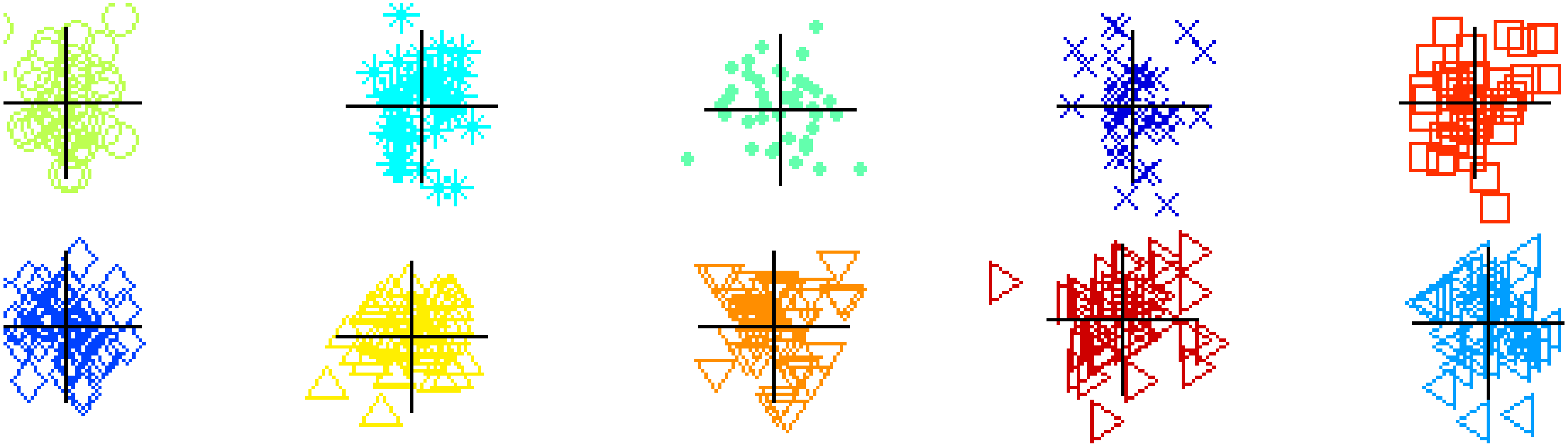
Bisecting K-means algorithm

- Variant of K-means that can produce a partitional or a hierarchical clustering

1. Initialize the list of clusters to contain the cluster containing all points.
2. **repeat**
3. Select a cluster from the list of clusters
4. **for** $i = 1$ to *number_of_iterations* **do**
5. Bisect the selected cluster using basic K-means
6. **end for**
7. Add the two clusters from the bisection with the lowest SSE to the list of clusters.
8. **until** Until the list of clusters contains K clusters

CLUTO: http://glaros.dtc.umn.edu/gkhome/cluto/cluto/overview

Bisecting K-means Example



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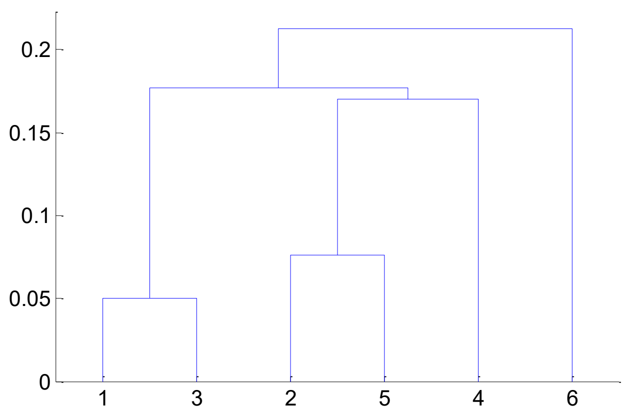
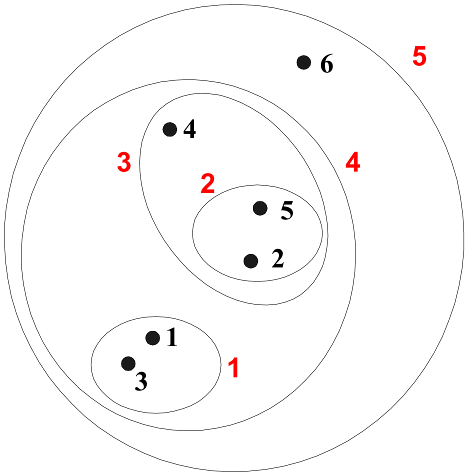
Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
  - A tree like diagram that records the sequences of merges or splits





Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
  - Any desired number of clusters can be obtained by ‘cutting’ the dendrogram at the proper level

- They may correspond to meaningful taxonomies
  - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, …)

Hierarchical Clustering

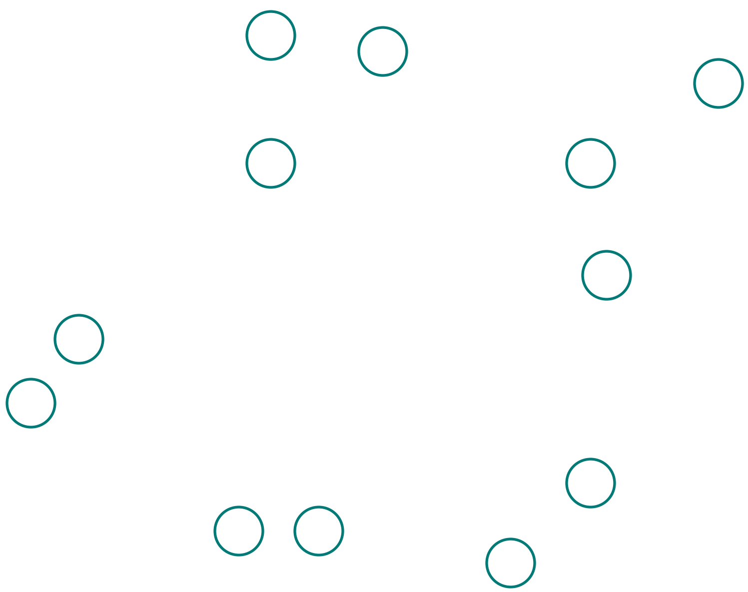
- Two main types of hierarchical clustering
  - Agglomerative:
    - Start with the points as individual clusters
    - At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
  - Divisive:
    - Start with one, all-inclusive cluster
    - At each step, split a cluster until each cluster contains an individual point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
  - Merge or split one cluster at a time

Agglomerative Clustering Algorithm

- Most popular hierarchical clustering technique
- Basic algorithm is straightforward
  1. Compute the proximity matrix
  2. Let each data point be a cluster
  3. **Repeat**
  4. Merge the two closest clusters
  5. Update the proximity matrix
  6. **Until** only a single cluster remains
- Key operation is the computation of the proximity of two clusters
  - Different approaches to defining the distance between clusters distinguish the different algorithms

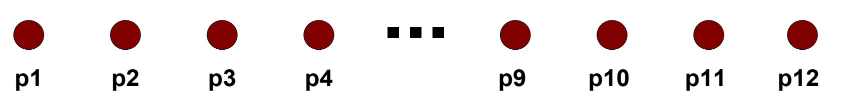
Starting Situation

- Start with clusters of individual points and a proximity matrix



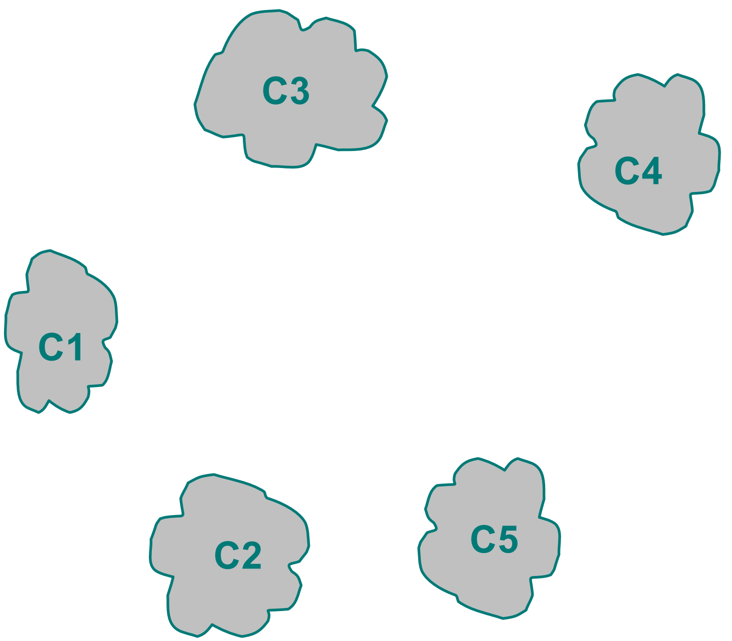
	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix



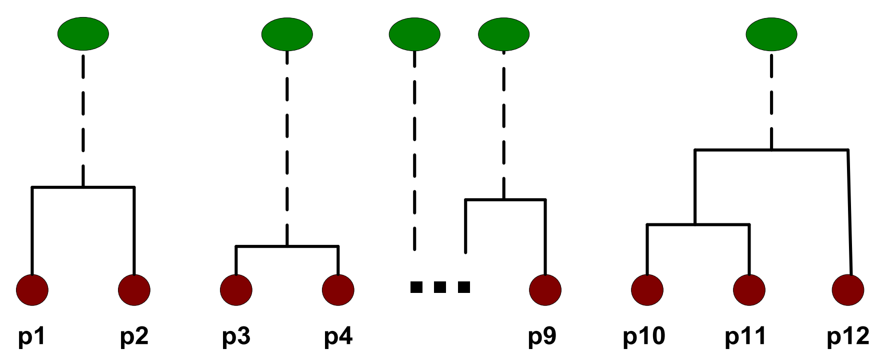
Intermediate Situation

- After some merging steps, we have some clusters



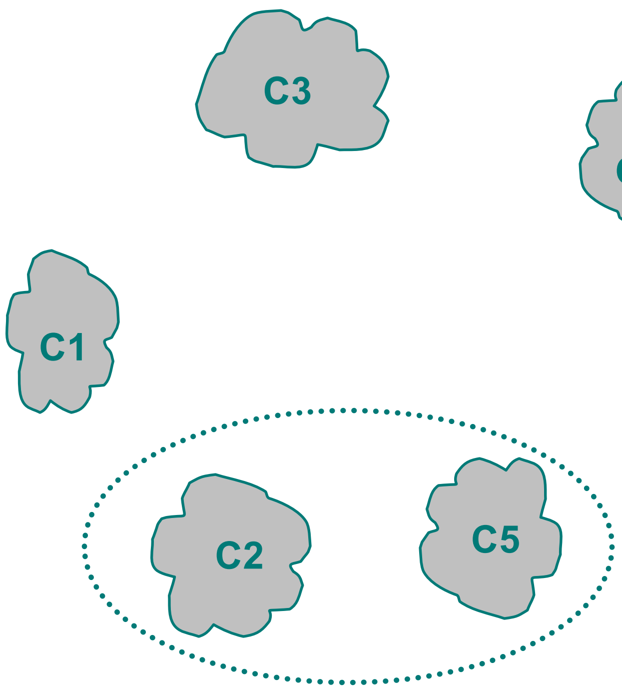
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

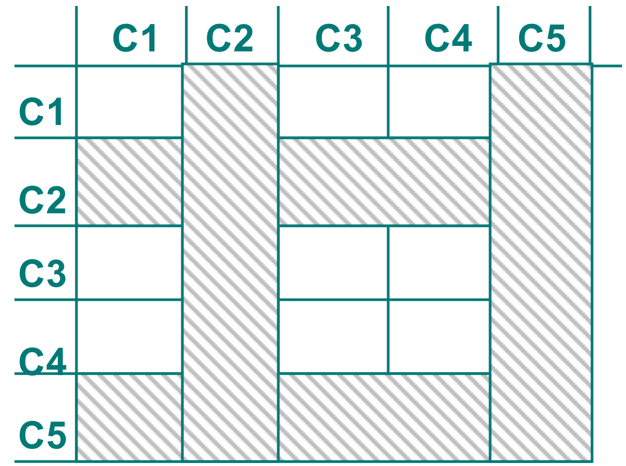
Proximity Matrix



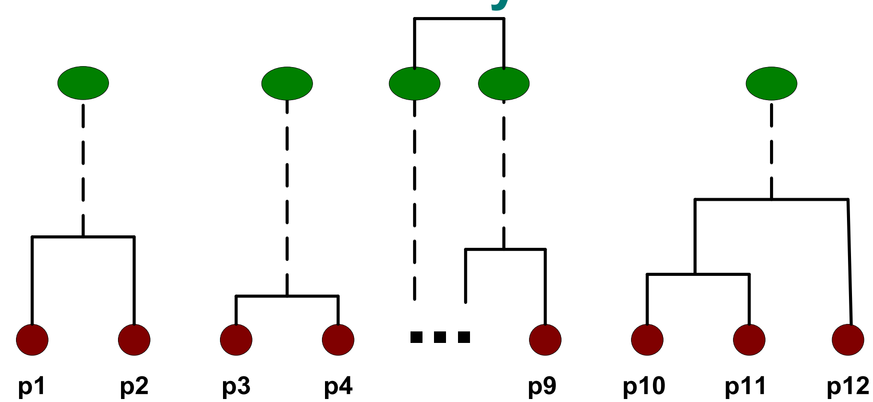
Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.



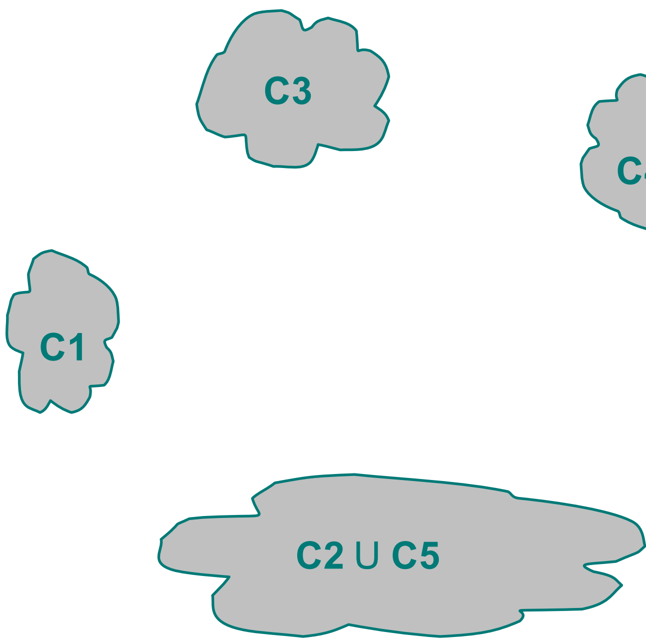


Proximity Matrix



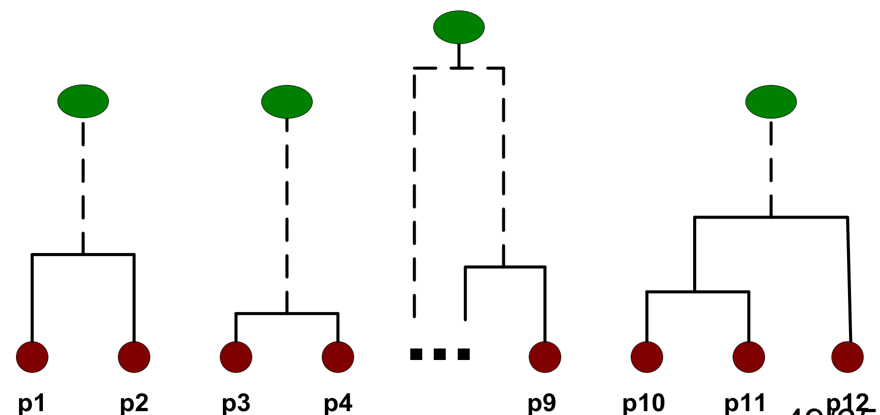
After Merging

- The question is “How do we update the proximity matrix?”

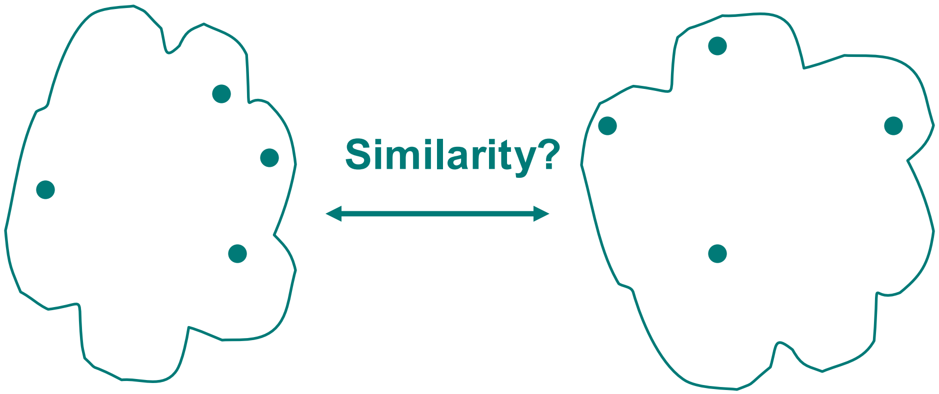


	C1	C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



How to Define Inter-Cluster Distance

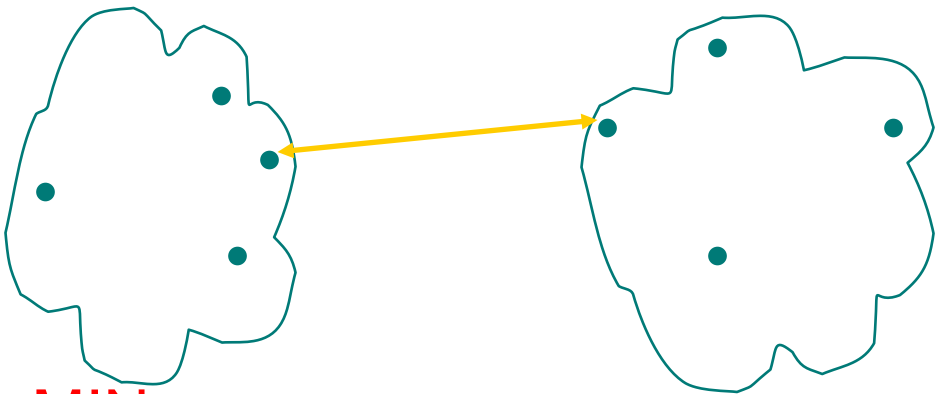


	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
. . .						

Proximity Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  – Ward's Method uses squared error

How to Define Inter-Cluster Similarity



	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

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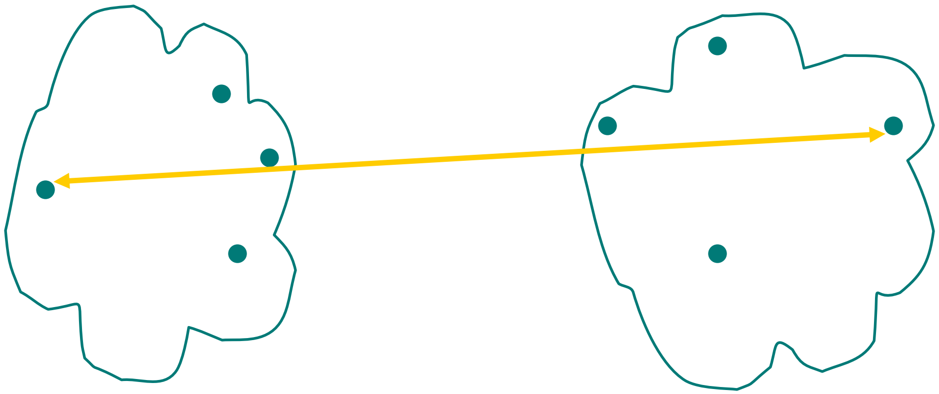
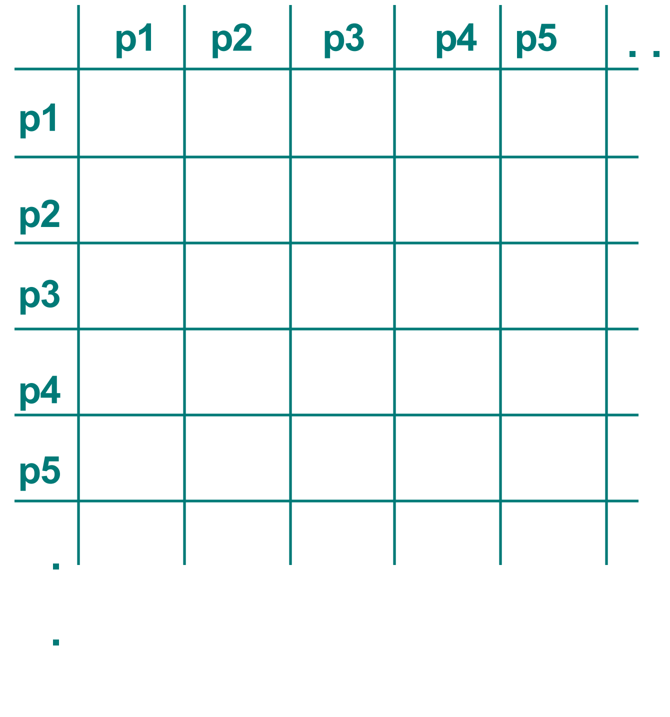
How to Define Inter-Cluster Similarity





Proximity Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
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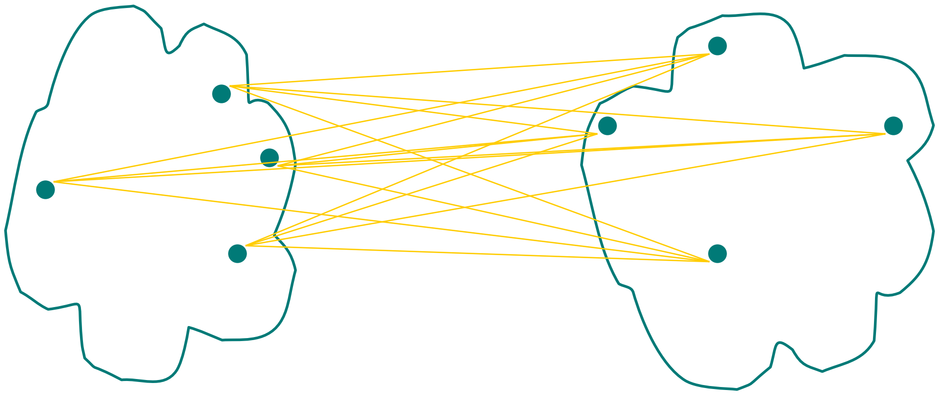
How to Define Inter-Cluster Similarity

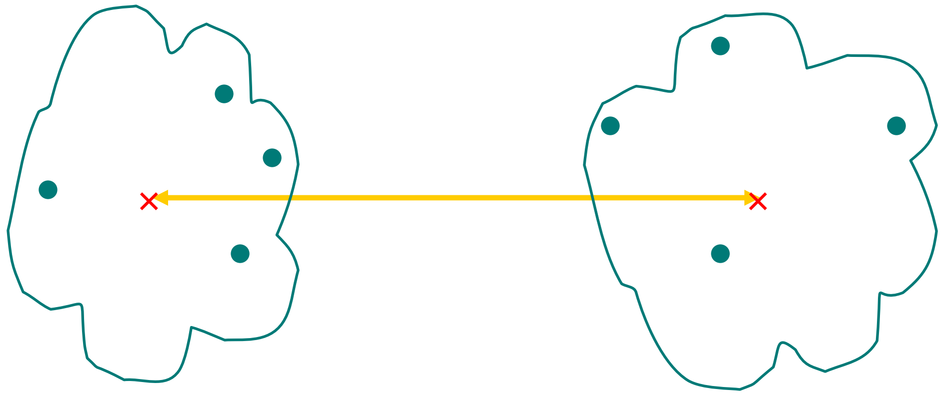


	p1	p2	p3	p4	p5	. . .
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  – Ward's Method uses squared error

How to Define Inter-Cluster Similarity



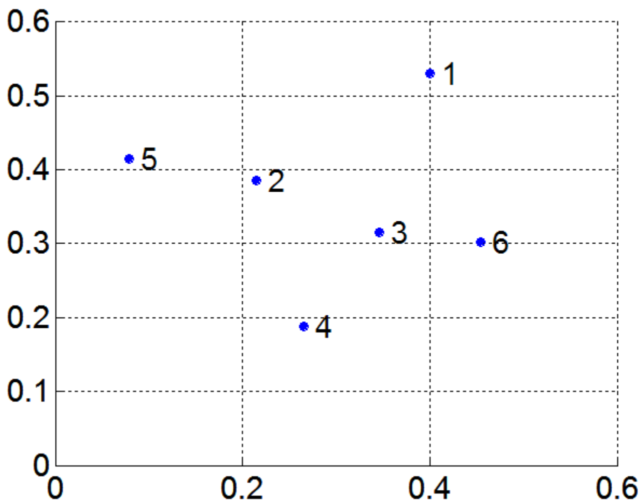
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  – Ward's Method uses squared error

MIN or Single Link

- Proximity of two clusters is based on the two closest points in the different clusters
  - Determined by one pair of points, i.e., by one link in the proximity graph
- Example:



Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

MIN or Single Link

- Proximity of two clusters is based on the two closest points in the different clusters
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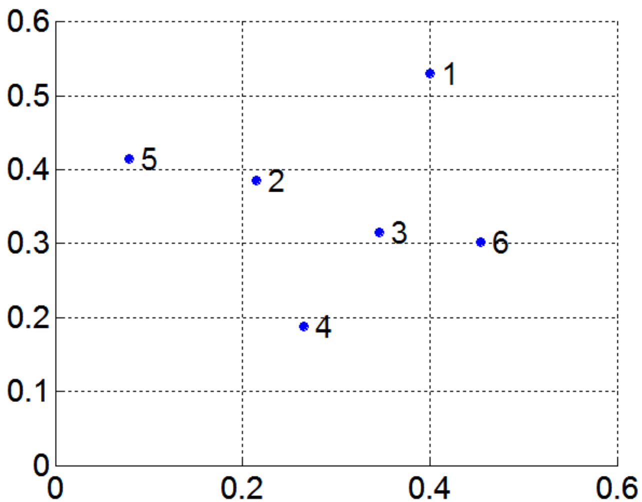
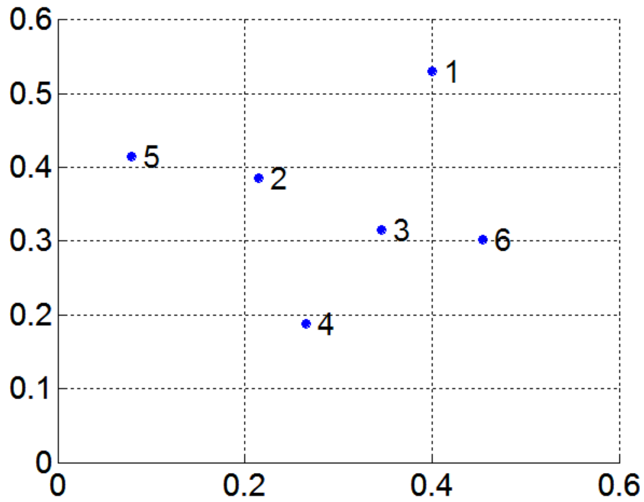
- Example:



Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

MIN or Single Link

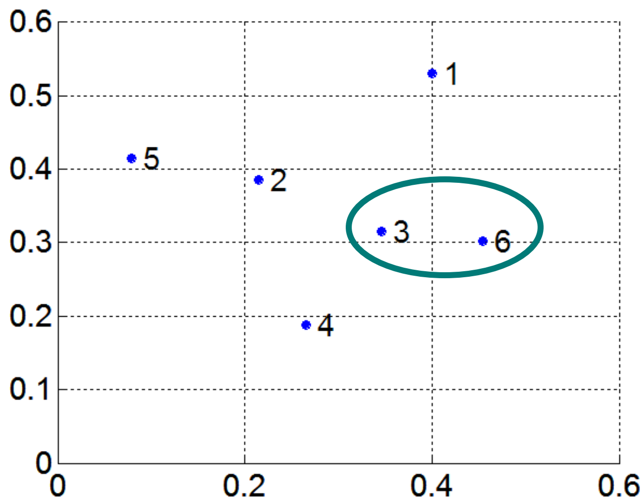


Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

P3,P6 are closest to each other, merge as first level

MIN or Single Link



Distance Matrix:

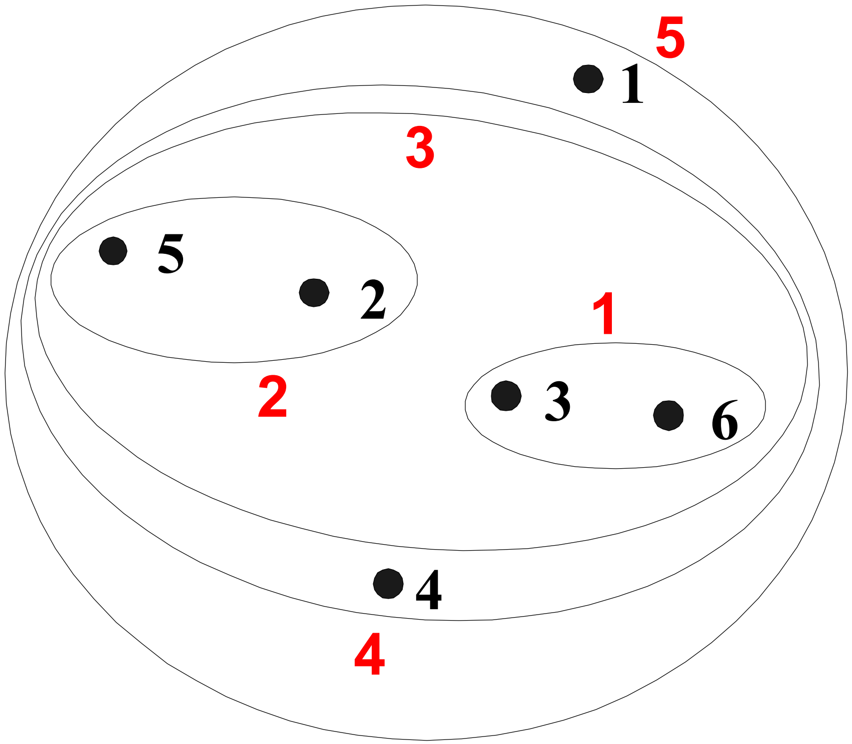
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

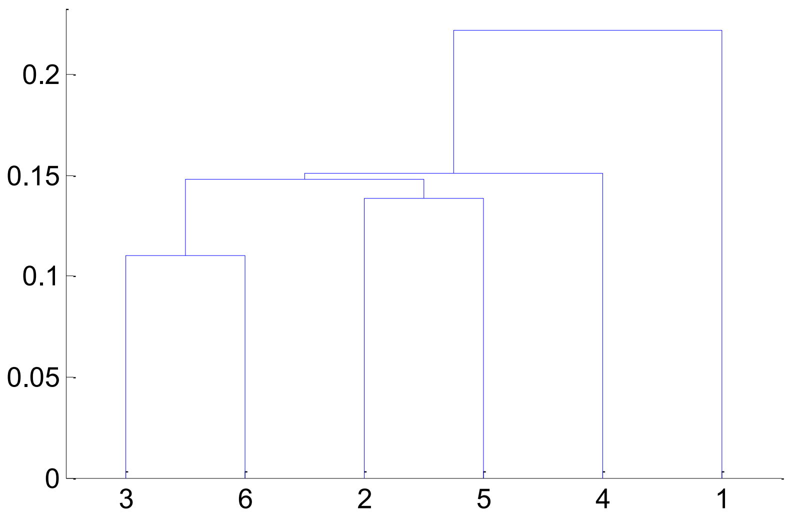
minimum

Update

	P1	P2	P3&P6	P4	P5
P1	0	0.24	0.22	0.37	0.34
P2		0	0.15	0.2	0.14
P3&P6			0	0.15	0.28
P4				0	0.29
P5					0

Hierarchical Clustering: MIN



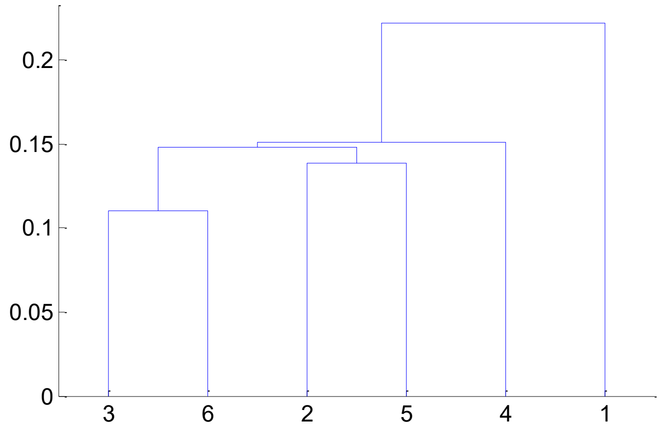


Nested Clusters

Dendrogram

In-class Quiz

1. Compute the distance matrix after merging Point 2 and Point 5.
2. According to the new distance matrix, which point (cluster) will be merged next?



	P1	**P2**	**P3&P6**	**P4**	**P5**
P1	0	0.24	0.22	0.37	0.34
P2		0	0.15	0.2	0.14
P3&P6			0	0.15	0.28
P4				0	0.29
P5					0

IS6400 Business Data Analytics

Dr. LIU Junming

City University of Hong Kong

Lecture 5

- Quick Review
- Introduction to Clustering
- Kmeans Clustering
- Hierarchical Clustering
- **Density-based Clustering**

Reading Material:

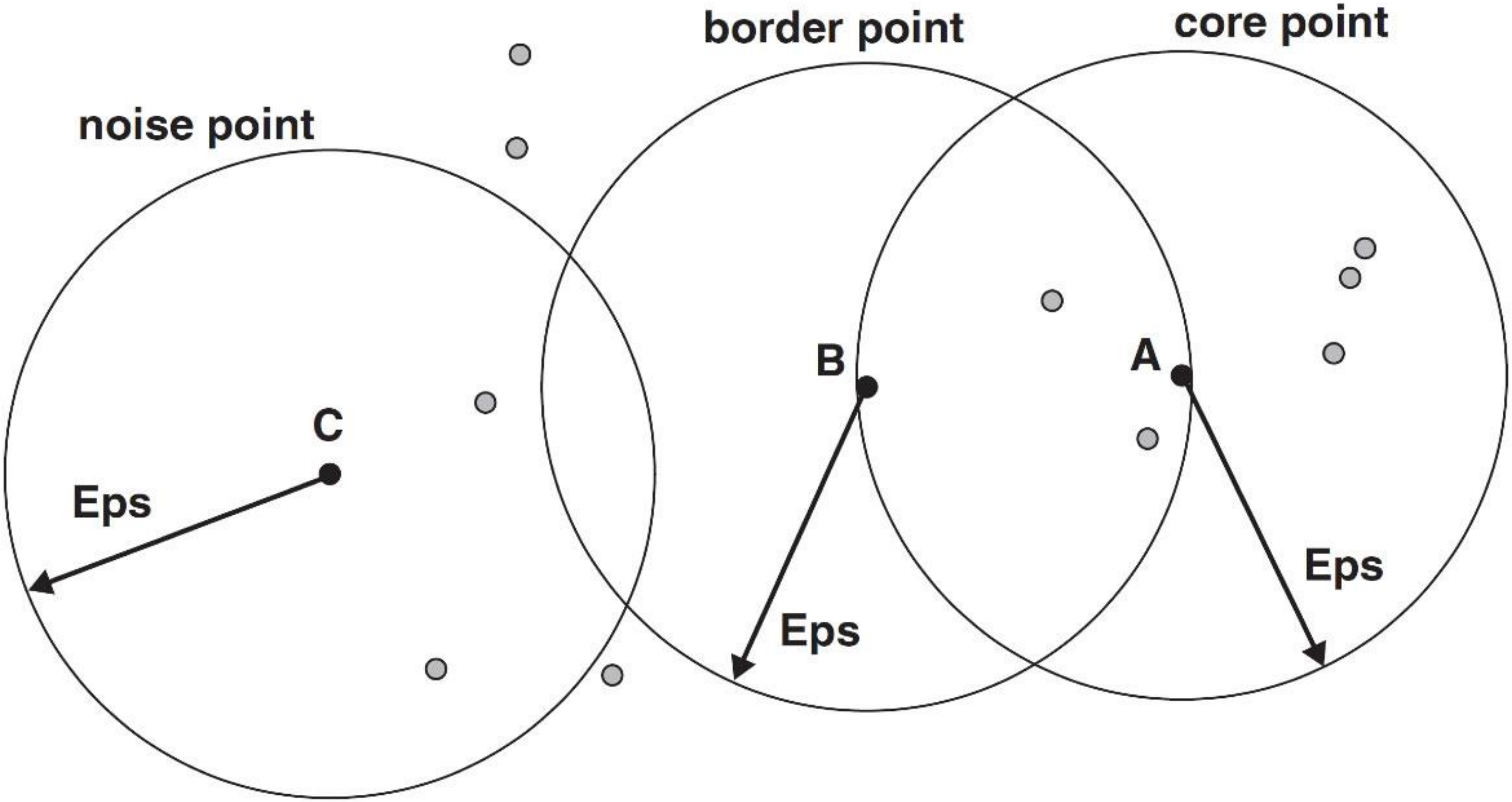
- DM Chapter 7,8

DBSCAN

- DBSCAN is a density-based algorithm.
  - Density = number of points within a specified radius (Eps)
  - A point is a **core point** if it has at least a specified number of points (MinPts) within Eps
    - These are points that are at the interior of a cluster
    - Counts the point itself
  - A **border point** is not a core point, but is in the neighborhood of a core point
  - A **noise point** is any point that is not a core point or a border point

DBSCAN: Core, Border, and Noise Points

MinPts = 7



DBSCAN Algorithm

- Eliminate noise points
- Perform clustering on the remaining points

$current\_cluster\_label \leftarrow 1$

for all core points **do**

  if the core point has no cluster label **then**

    $current\_cluster\_label \leftarrow current\_cluster\_label + 1$

    Label the current core point with cluster label $current\_cluster\_label$

  end if

  for all points in the Eps-neighborhood, except i^{th} the point itself **do**

    if the point does not have a cluster label **then**

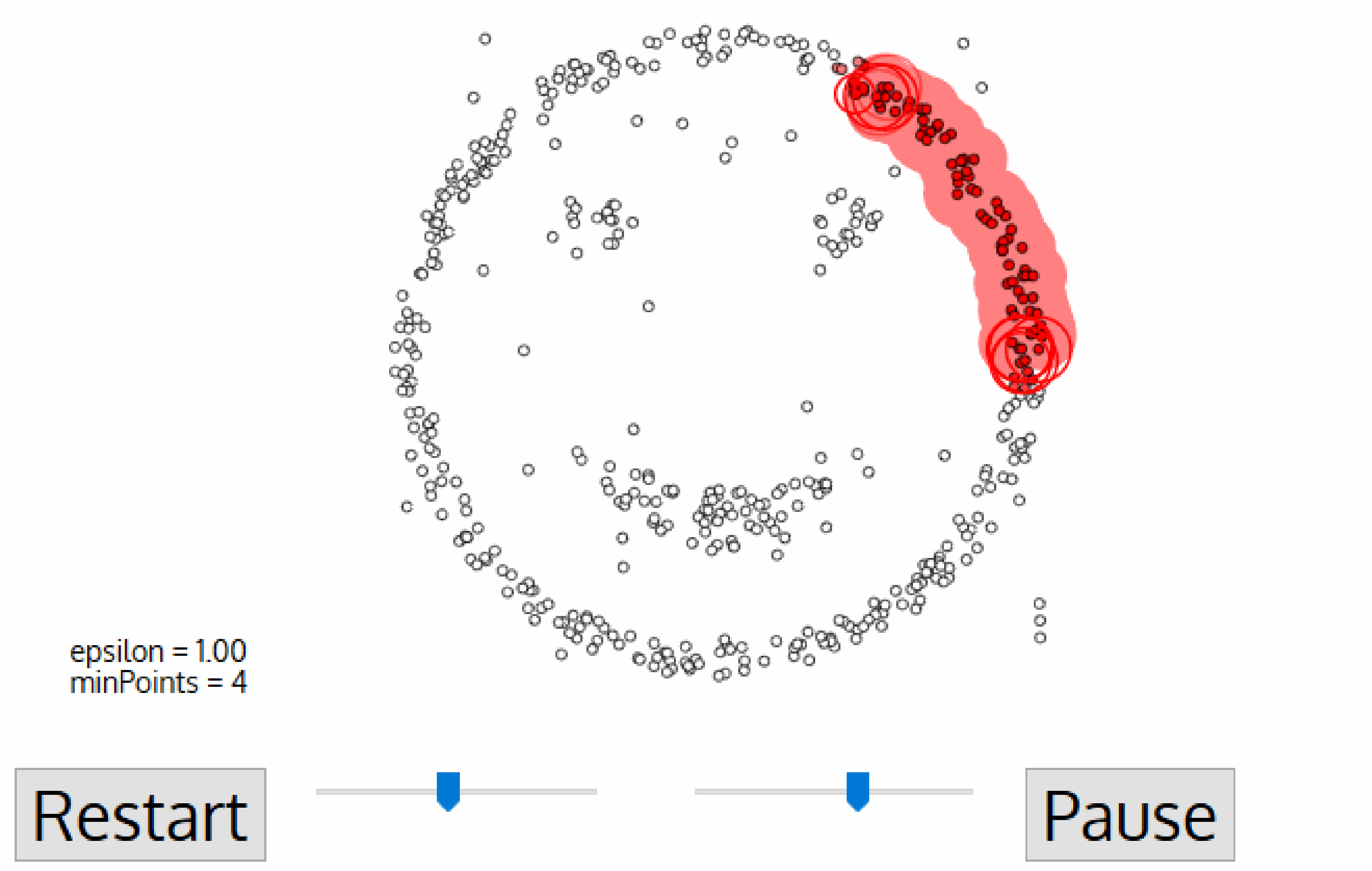
      Label the point with cluster label $current\_cluster\_label$

    end if

  end for

end for

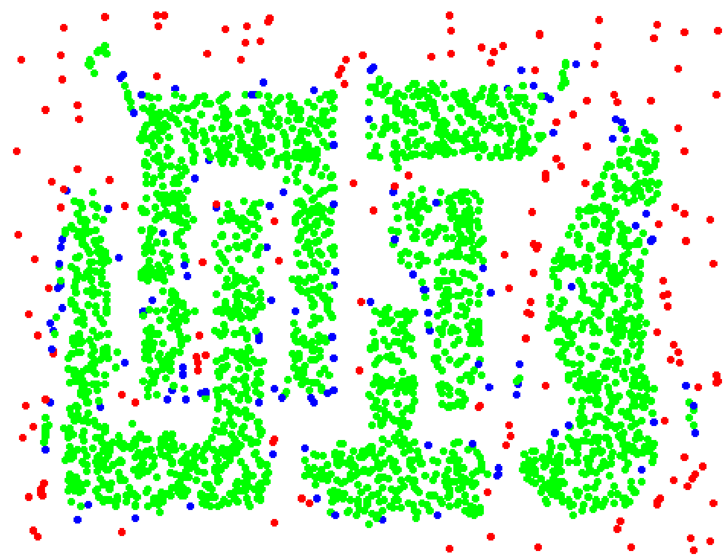
DBSCAN Process



REF: https://www.kdnuggets.com/2020/04/dbscan-clustering-algorithm-machine-learning.html

DBSCAN: Core, Border and Noise Points

Original Points

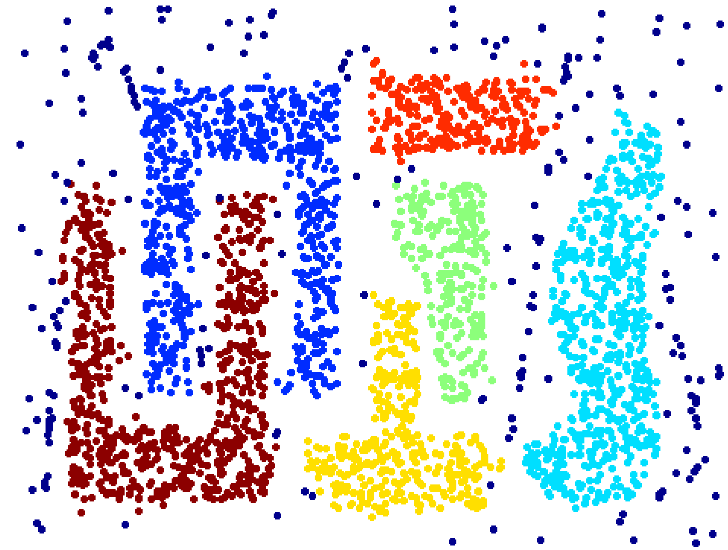


Point types: core, border and noise

Eps = 10, MinPts = 4

When DBSCAN Works Well

Original Points



Clusters

- Resistant to Noise
- Can handle clusters of different shapes and sizes